

MODULE 3

OBJECT-ORIENTED PROGRAMMING IN JAVA

From core OOP principles to SOLID, design patterns, and real domain modeling.







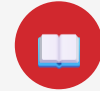
OBJECT-ORIENTED PROGRAMMING IN JAVA

Module goal, what you will learn, why it matters, and the key takeaways ahead. (1 of 5)



MODULE GOAL

- Understand and apply Object-Oriented Programming (OOP) concepts in Java to create reusable, modular, and maintainable programs.



WHAT YOU WILL LEARN

- Classes & Objects, Constructors
- Encapsulation, Inheritance
- Polymorphism, Abstraction
- Real-world OOP examples



WHY IT MATTERS

- Models real-world entities
- Improves code reusability
- Enhances maintainability
- Supports scalability and teamwork
- Foundation for frameworks and large applications



KEY TAKEAWAYS

- Think in terms of objects and classes
- Follow OOP principles for clean code
- Use inheritance and polymorphism for flexibility
- Encapsulate data for security
- Design for reuse and extension

KEY TAKEAWAY Mastering OOP is the key to building powerful, scalable, professional Java applications.

OBJECT ORIENTED PROGRAMMING (OOP) IN JAVA

OOP is a programming paradigm that uses objects and classes to design applications that are modular, reusable and easy to maintain.

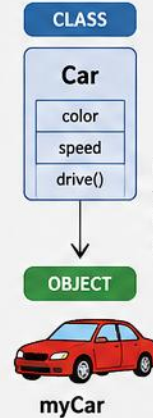
1. CLASS & OBJECT

A Class is a blueprint for creating objects. An Object is an instance of a class.

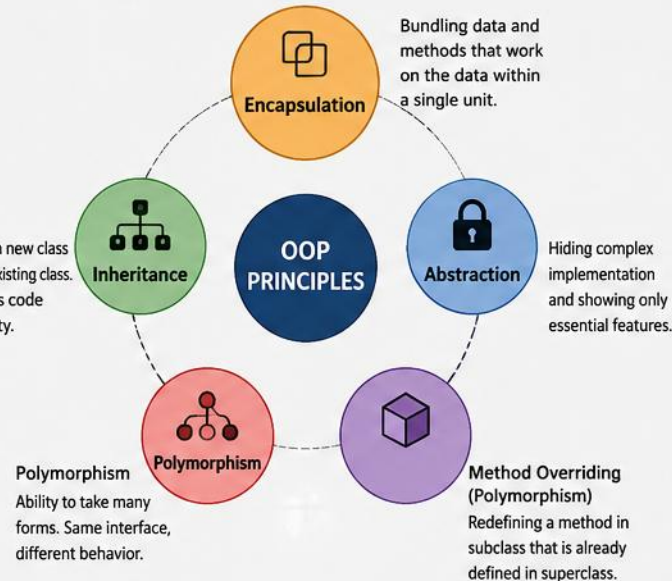
```
class Car {
    String color;
    int speed;

    void drive() {
        System.out.println("Driving");
    }
}

Car myCar = new Car(); // Object
myCar.color = "Red";
myCar.speed = 120;
```



2. KEY OOP PRINCIPLES

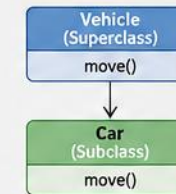


3. INHERITANCE EXAMPLE

```
class Vehicle {
    void move() {
        System.out.println("Moving");
    }
}

class Car extends Vehicle {
    void move() {
        System.out.println("Car is moving");
    }
}

Car c = new Car();
c.move(); // Output: Car is moving
```

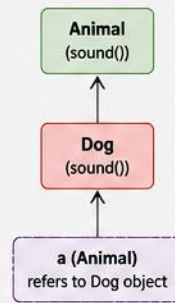


4. POLYMORPHISM EXAMPLE

```
class Animal {
    void sound() {
        System.out.println("Some sound");
    }
}

class Dog extends Animal {
    void sound() {
        System.out.println("Bark");
    }
}

Animal a = new Dog(); // Upcasting
a.sound(); // Output: Bark (runtime polymorphism)
```

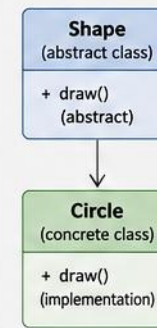


5. ABSTRACTION EXAMPLE

```
abstract class Shape {
    abstract void draw(); // abstract method
}

class Circle extends Shape {
    void draw() {
        System.out.println("Drawing Circle");
    }
}

Shape s = new Circle();
s.draw(); // Output: Drawing Circle
```



6. BENEFITS OF OOP

	Modularity Breaks complex programs into smaller, manageable pieces.
	Reusability Code can be reused through inheritance and composition.
	Maintainability Easy to update and modify existing code.
	Scalability Easier to extend and build large applications.



OOP CONCEPTS AT A GLANCE

The six ideas behind everything in this module, and the flow you'll follow to apply them. (2 of 5)



Class

A blueprint or template for creating objects. Defines attributes and behaviors.



Object

An instance of a class. Has state (data) and behavior (methods).



Encapsulation

Bundling data and methods together, restricting direct access to some components.



Inheritance

Creating a new class from an existing class to reuse code and extend functionality.



Polymorphism

One interface, multiple implementations. Same method calls behave differently.



Abstraction

Showing only essential features and hiding the implementation details.



PROGRAM FLOW

- 1 Define Class (Attributes + Methods)
- 2 Create Objects (Instantiate the class)
- 3 Use Constructors (Initialize object state)
- 4 Access & Modify Data (Using methods)
- 5 Apply OOP Principles (Inheritance, Polymorphism, etc.)
- 6 Build Real-World Programs (Reusable and Maintainable)

KEY TAKEAWAY Six ideas, one flow: define, create, construct, use, apply principles, and build real programs.

SAMPLE PROGRAM & REAL-WORLD ANALOGY

A first, complete look at classes, constructors, and methods working together. (3 of 5)

```
class Person {  
    private String name; private int age;  
  
    public Person(String name, int age) {  
        this.name = name; this.age = age;  
    }  
  
    public void introduce() {  
        System.out.println("Hi, I am " + name +  
            " and I am " + age + " years old.");  
    }  
    public int getAge() { return age; }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        Person p1 = new Person("Alice", 21);  
        p1.introduce();  
        System.out.println("Age: " + p1.getAge());  
    }  
}
```



SAMPLE OUTPUT

```
Hi, I am Alice and I am 21 years old.  
Age: 21
```



REAL-WORLD ANALOGY

- Class: Blueprint of a Car
- Object: A specific Car (e.g., my car)
- Encapsulation: Engine details hidden inside the car
- Inheritance: ElectricCar is a Car
- Polymorphism: Different cars start() differently
- Abstraction: Driver uses steering without knowing internal mechanics

KEY TAKEAWAY Classes define the blueprint; constructors initialize state; methods define behavior.

LAB HIGHLIGHTS, DELIVERABLES & EVALUATION

What you'll build for this module, what to submit, and how it's graded. (4 of 5)

LAB HIGHLIGHTS

- Build a menu-driven Banking Management System
- Model an abstract Account with SavingsAccount and CurrentAccount
- Implement the Printable interface
- Track deposits and withdrawals with Transaction records
- Coordinate operations through a BankService layer
- Demonstrate polymorphism via Account[] display
- Apply SOLID principles across the design

DELIVERABLES

- Java source code files (.java) — com.academy.bank package
- Console output screenshots (create, deposit, withdraw, display)
- UML class diagram (Account hierarchy + interface)
- Short design note (SOLID + inheritance/polymorphism)
- Documented compile/run commands (javac -d out / java -cp out)

CRITERIA	WEIGHT
Class Design & Encapsulation	20%
Inheritance, Abstraction & Interface	25%
Polymorphism & Service Operations	25%
Menu App, Packaging & Compile	15%
UML, Code Quality & Evidence	15%
Total	100%

KEY TAKEAWAY Build the Banking Management System lab, applying every OOP concept from this module.

TIPS FOR SUCCESS & WHAT'S NEXT

Practical advice for this module, and a preview of what's coming next. (5 of 5)

★ TIPS FOR SUCCESS

- Start with simple classes and build gradually.
- Think about real-world entities.
- Keep data private and access through methods.
- Use meaningful names.
- Test each part as you build.

➔ NEXT STEP

In the next module, we will explore Exception Handling and File I/O to make our programs more robust and data-aware.

"Mastering OOP is the key to building powerful, scalable, and professional Java applications."

Keep Practicing, Keep Building! 🏆

KEY TAKEAWAY Mastering OOP is the key to building powerful, scalable, and professional Java applications.

WHY OOP MATTERS IN ENTERPRISE DEVELOPMENT

OOP helps teams build software that is easier to build, maintain, extend, and scale. (1 of 2)

✗ WITHOUT OOP

- Code becomes hard to manage and change
- High coupling and duplication
- Difficult to test and maintain
- Scaling and adding features becomes risky and expensive

✓ WITH OOP

- Modular, organized, and reusable code
- Easier to extend and evolve
- Lower maintenance cost
- Better quality and reliability
- Supports long-term growth

💡 KEY IDEA

Enterprise systems are complex, evolve constantly, and are built by large teams over years. OOP provides proven principles to manage this complexity effectively.

KEY TAKEAWAY OOP is not just a coding style — it's a foundation for building robust, future-ready applications.

Why OOP Matters in Enterprise Development

OOP principles help enterprise applications handle complexity, evolve with change, and deliver business value at scale.



OOP is not just a coding style—it's a foundation for building robust, maintainable, and future-ready enterprise solutions.



ENTERPRISE

HOW OOP PRINCIPLES DELIVER VALUE IN ENTERPRISE DEVELOPMENT

Six ways OOP pays off in real enterprise development. (2 of 2)



Modularity & Reusability

Classes and objects break the system into smaller, reusable components. Reuse saves time and reduces defects. Example: a common Logger or DatabaseConnector class reused across many modules.



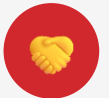
Maintainability

Encapsulation hides internal details and localizes changes. Easier to understand, debug, and maintain over time. Example: changing how data is stored doesn't affect other parts of the system.



Extensibility

Inheritance and polymorphism let you extend behavior without modifying existing code. Example: add a new payment method by creating a new class without changing existing code.



Supports Team Collaboration

Clear class responsibilities and interfaces enable multiple developers to work in parallel with minimal conflicts. Example: different teams work on different classes/modules independently.



Scalability

OOP designs are easier to scale as the application grows in size, users, and business complexity. Example: add new features or modules without breaking existing functionality.



Reliability & Security

Encapsulation and abstraction protect critical data and enforce valid usage, reducing errors and security risks. Example: private fields and controlled access ensure data integrity.

Impact on the Business: 🚀 **Faster time to market** 💰 **Lower total cost of ownership** ★ **Higher quality software** 🔗 **Easier integration with other systems** 🔄 **Better adaptability to change**

KEY TAKEAWAY OOP is a foundation for building robust, flexible, future-ready applications — it empowers teams to deliver value today and evolve confidently for tomorrow.

LEARNING OBJECTIVES

By the end of this module, you will be able to design, implement, and use object-oriented concepts to build maintainable and extensible Java programs. (1 of 4)



1

Understand Classes and Objects

Define classes as blueprints and create objects as instances with state and behavior.



2

Use Constructors Effectively

Create and use constructors to initialize objects and understand constructor overloading.



3

Apply Encapsulation

Use access modifiers to protect data and provide controlled access through getters and setters.



4

Implement Inheritance

Create new classes by extending existing classes to reuse and extend functionality.



5

Leverage Polymorphism

Understand method overriding and use polymorphism to write flexible and extensible code.

KEY TAKEAWAY In this module, you will build ten core capabilities — see part 2 of 4 for the rest.

LEARNING OBJECTIVES (CONTINUED)

(2 of 4)



6

Achieve Abstraction

Define abstract classes and interfaces to focus on essential behavior and hide implementation details.



7

Understand Relationships Between Classes

Recognize and model relationships: Association, Aggregation, Composition.



8

Design Real-World OOP Models

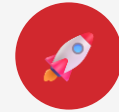
Analyze real-world scenarios and design class diagrams to model entities and their interactions.



9

Write Clean and Maintainable Code

Follow OOP best practices: Naming, Structure, Cohesion, Low coupling.



10

Build OOP Programs

Develop complete Java programs using OOP concepts to solve practical problems.

KEY TAKEAWAY Objectives 7 and 9 cover class relationships and clean-code practices in more depth ahead.

YOU WILL WORK WITH

The building blocks you'll use, and the systems you'll practice on. (3 of 4)

YOU WILL WORK WITH

- Classes, objects, and constructors
- Access modifiers and encapsulation
- Inheritance hierarchies
- Interfaces and abstract classes
- Polymorphism and method overriding
- UML class diagrams
- OOP best practices and design principles



Vehicle Management System

Model a Car system using classes and inheritance:

- Vehicle
- Car
- Truck
- Bike



Banking System

Encapsulate BankAccount data and expose safe operations:

- deposit()
- withdraw()
- getBalance()



Payment System

Use interfaces to support multiple payment types via polymorphism:

- CreditCard
- PayPal
- UPI

KEY TAKEAWAY Example outcomes — Vehicle Management, Banking, and Payment systems — put every concept into practice.

SUCCESS CRITERIA

By the end of this module, you should be able to check off every one of these. (4 of 4)



I can define classes and create objects.



I can apply encapsulation to protect data.



I can use inheritance and polymorphism correctly.



I can design class diagrams for real-world problems.



I can build and run OOP-based programs.

 **GOAL** Build a strong OOP foundation to write flexible, reusable, and enterprise-ready Java applications.

KEY TAKEAWAY Goal: build a strong OOP foundation to write flexible, reusable, enterprise-ready Java applications.

BUSINESS SCENARIO: BANKING MANAGEMENT SYSTEM

We will use a real-world banking scenario to understand and apply OOP concepts in Java. (1 of 3)

SCENARIO OVERVIEW

A bank wants a system to manage its customers, accounts, and transactions. The system should be:

- Secure
- Easy to maintain
- Extendable to support new types of accounts and services

MAIN ENTITIES IN THE SYSTEM



Customer

Represents a bank customer. Has personal details and one or more accounts.



Account

Abstract concept for all types of accounts. Has account number, balance, and common operations.



Transaction

Represents a transaction on an account. Stores type, amount, date, and description.

KEY REQUIREMENTS

- Customers can have one or more accounts
- Support Savings, Current, and Fixed Deposit accounts
- Perform deposits, withdrawals, and balance inquiries
- Maintain transaction history for each account
- Ensure data security through encapsulation
- Allow future extensions (e.g., Loan Account)

KEY TAKEAWAY A bank needs a system that is secure, easy to maintain, and extendable to new account types.

OOP CONCEPTS APPLIED

Every pillar of OOP maps directly onto the banking scenario. (2 of 3)



Encapsulation

Protect account details (balance, account number) using private fields and public methods.



Inheritance

Account is the base class. Derived classes include SavingsAccount, CurrentAccount, FixedDepositAccount.



Polymorphism

Different account types override methods such as calculateInterest() and applyCharges().



Abstraction

Define abstract methods in the Account class that must be implemented by subclasses.



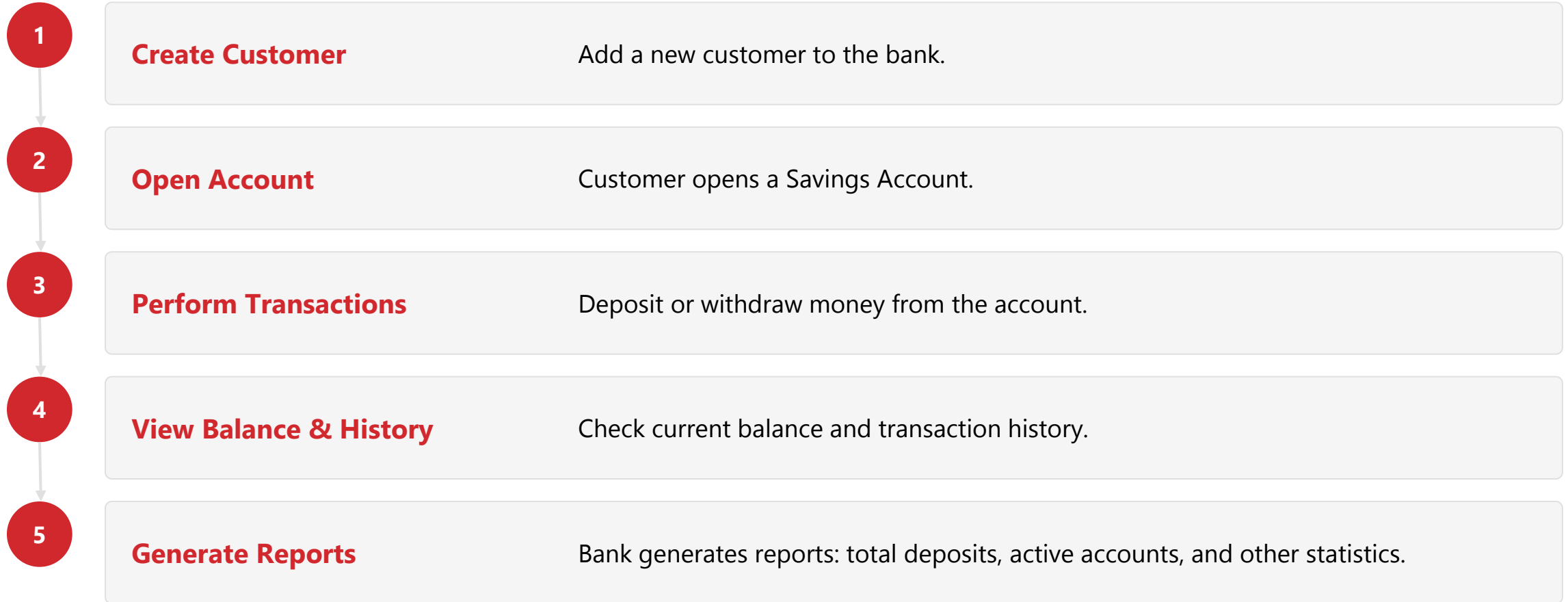
Association

Customer has Accounts.
Account has Transactions.

KEY TAKEAWAY Encapsulation, Inheritance, Polymorphism, Abstraction, and Association — all five map onto one real system.

EXAMPLE USE CASE FLOW

From onboarding a customer to generating reports. (3 of 3)



KEY TAKEAWAY Learning Outcome: design and implement a modular, reusable, and extensible banking system using core OOP principles.

WHAT IS OBJECT-ORIENTED PROGRAMMING?

OOP organizes software design around data, or objects, rather than functions and logic alone. (1 of 3)

KEY IDEA

OOP models real-world entities as objects that have:

- State — attributes / data
- Behavior — methods / actions
- Identity — unique reference

It helps us build software that is:

- **Modular**
- **Reusable**
- **Maintainable**
- **Easier to extend**

REAL-WORLD ANALOGY — THINK OF A CAR

STATE (DATA)

Make: Toyota
Model: Camry
Color: Red
Speed: 60 km/h

BEHAVIOR (METHODS)

start()
accelerate()
brake()
stop()

IDENTITY — each car has a unique identity:

- VIN (Vehicle Identification Number)
- Registration Number

Object-Oriented Programming (OOP) is a programming paradigm that organizes software design around data, or objects, rather than functions and logic alone.

KEY TAKEAWAY Every object has State, Behavior, and Identity — the three ingredients OOP is built on.

OBJECT ORIENTED PROGRAMMING (OOP)

A programming paradigm that organizes code around objects and classes rather than functions and logic.

KEY PRINCIPLES OF OOP



1. CLASS & OBJECT

Class is a blueprint.
Object is an instance of a class.



2. ENCAPSULATION

Binding data (fields) and methods together and restricting direct access to some of the object's components.



3. INHERITANCE

Deriving a new class from an existing class to reuse code and extend functionality.



4. POLYMORPHISM

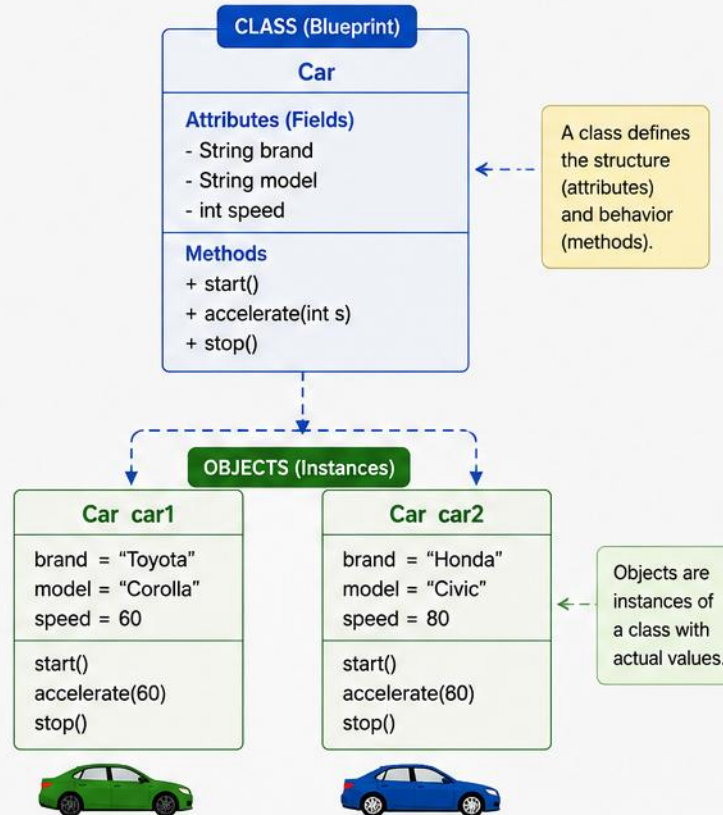
One interface, multiple implementations.
The same method behaves differently for different objects.



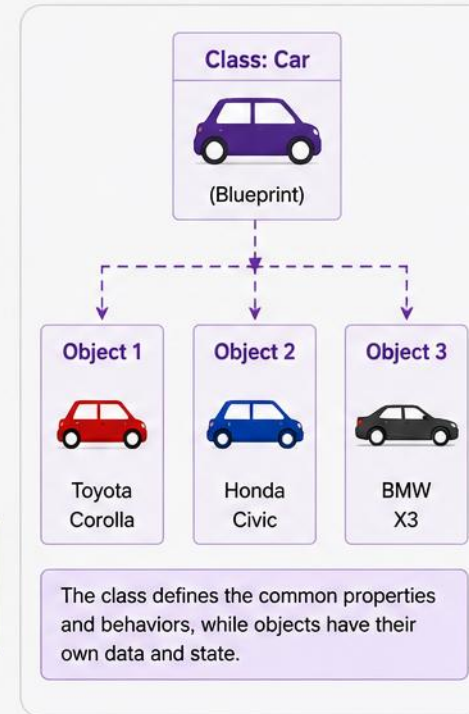
5. ABSTRACTION

Hiding complex implementation and showing only essential features to the user.

CLASS VS OBJECT



REAL WORLD ANALOGY



WHY OOP?



- ✓ Improves code reusability
- ✓ Enhances maintainability
- ✓ Supports scalability
- ✓ Models real world entities
- ✓ Promotes clean and organized code



OOP IN JAVA

WHY OOP?

Five reasons OOP has become the standard way to build serious software. (2 of 3)



Models real-world entities naturally

Easy to understand and relate.



Promotes reusability

Write once, reuse many times.



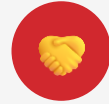
Improves maintainability

Changes in one part have less impact on others.



Enhances scalability

Easy to add new features and handle growing complexity.



Supports teamwork

Code is organized into well-defined classes with clear responsibilities.

★ BENEFITS OF OOP

Modular Design • Reusability of Code • Data Security and Integrity • Easier Maintenance and Extension • Better for Large and Complex Applications

KEY TAKEAWAY OOP is not just a way to write code — it's a way to think about problems.

THE FOUR PILLARS OF OOP

Every OOP concept in Java builds on these four ideas. (3 of 3)



Encapsulation

Bundling data and methods together in a class and restricting direct access to some components.

EXAMPLE

Account balance is hidden (private) and can be accessed only through methods like `getBalance()`.



Inheritance

Creating a new class from an existing class to reuse and extend its functionality.

EXAMPLE

`SavingsAccount` is-a `Account` and inherits its properties and methods.



Polymorphism

One interface, multiple implementations. The same method call can behave differently for different objects.

EXAMPLE

`calculateInterest()` works differently for `SavingsAccount` and `FixedDepositAccount`.



Abstraction

Hiding internal implementation details and showing only the essential features.

EXAMPLE

We use an `Account` object without worrying about how interest is actually calculated.

KEY TAKEAWAY OOP promotes clean design, code reuse, and easier software evolution — essential for robust enterprise applications.

CLASSES AND OBJECTS

In Java, everything is built around classes and objects. (1 of 3)

WHAT IS A CLASS?

A class is a blueprint or template that defines attributes (data/state) and methods (behavior/actions). Example: class Car defines what a car has and what it can do.

WHAT IS AN OBJECT?

An object is a real-world entity created from a class. An object has its own state (values) and can perform actions (methods). Example: myCar is an object of class Car.

REAL-WORLD ANALOGY

A class is like a blueprint of a house. Objects are the actual houses built from that blueprint.



**Blueprint
Class (Car)**



**House 1
myCar**



**House 2
yourCar**



**House 3
officeCar**

KEY TAKEAWAY

Classes help us design; objects help us represent and work with real-world entities in our programs.

KEY TAKEAWAY A class is a blueprint; an object is an instance of that class.

Classes and Objects in Java

A **Class** is a blueprint or template that defines the properties (fields) and behaviors (methods) of objects.
An **Object** is an instance of a class.



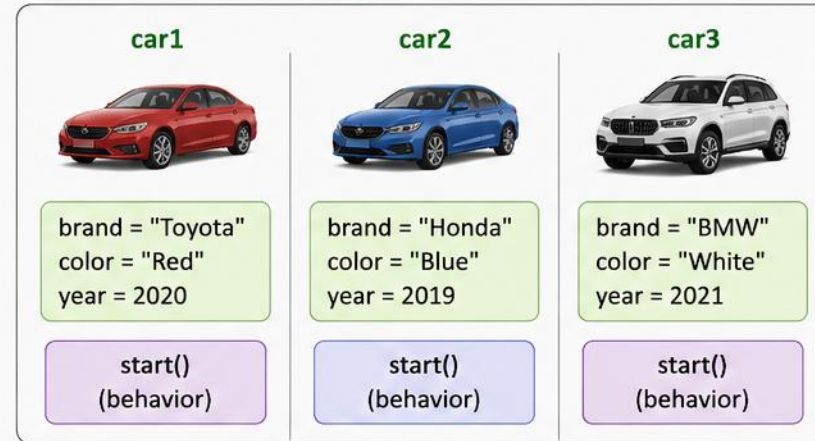
1. Class (Blueprint)

```
class Car {  
    // Fields (attributes)  
    String brand;  
    String color;  
    int year;  
  
    // Method (behavior)  
    void start() {  
        System.out.println("Car started");  
    }  
}
```



Objects are created from a class using the 'new' keyword.

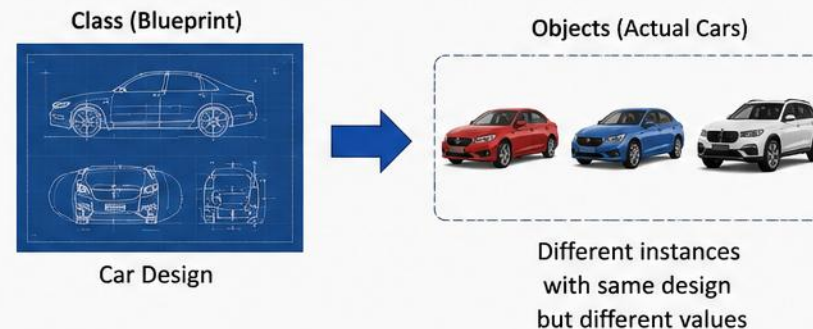
2. Objects (Instances)



3. Creating Objects in Java

```
// Creating objects  
Car car1 = new Car();  
Car car2 = new Car();  
Car car3 = new Car();  
  
// Setting values  
car1.brand = "Toyota"; car1.color = "Red"; car1.year = 2020;  
car2.brand = "Honda"; car2.color = "Blue"; car2.year = 2019;  
car3.brand = "BMW"; car3.color = "White"; car3.year = 2021;  
  
// Calling method  
car1.start(); // Output: Car started
```

4. Real-World Analogy



Key Takeaways

- Class defines what objects look like and what they can do.
- Objects are real entities created from a class.
- Multiple objects can be created from the same class, each with its own values.

Write Once, Run Anywhere

EXAMPLE IN JAVA

1. *Class Definition (Blueprint) — attributes and methods live together in one class. (2 of 3)*

```
class Car {  
  
    int speed;  
    String color;  
    String model;  
  
    // Method (behavior)  
    void start() {  
        System.out.println("Car is starting...");  
    }  
  
    void stop() {  
        System.out.println("Car is stopping...");  
    }  
}
```

ATTRIBUTES (STATE)

- speed
- color
- model

METHODS (BEHAVIOR)

- start()
- stop()

KEY TAKEAWAY A class bundles attributes (what an object has) and methods (what an object can do) into one blueprint.

EXAMPLE IN JAVA

2. Creating Objects — each object gets its own state, stored separately in memory. (3 of 3)

```
public class Main {  
    public static void main(String[] args) {  
  
        // Creating objects of class Car  
        Car myCar = new Car();  
        Car yourCar = new Car();  
        Car officeCar = new Car();  
  
        // Setting state  
        myCar.speed = 120;  
        myCar.color = "Red";  
        myCar.model = "Toyota Camry";  
  
        // Calling methods  
        myCar.start();  
        myCar.stop();  
    }  
}
```

MEMORY (RUNTIME)

myCar

Speed: 120, Color: Red, Model: Toyota Camry

yourCar

Speed: 80, Color: Blue, Model: Honda City

officeCar

Speed: 100, Color: White, Model: BMW 5 Series

KEY POINTS

- A class is a blueprint; an object is an instance of that class
- Multiple objects can be created from a single class
- Each object has its own state, stored in memory
- Methods define the behavior that objects can perform

KEY TAKEAWAY Each object has its own copy of data, stored in separate memory.

TRY IT YOURSELF — EXERCISE 1: IDENTIFY ENTITIES

Reinforces: turning a real-world domain into classes and objects, as just covered.



1

Goal

List entities, attributes, and one responsibility each from a banking domain



2

Deliverable

A short entity table in notes.md — no code required



3

Domain

Customer, Account, Transaction — the entities from this module's scenario



4

Key concept

Each entity becomes a class; attributes become fields; behavior becomes methods



5

Reference

exercises/exercise-01-domain-entities.md

TRY IT NOW Complete Exercise 1 before continuing — see [exercises/exercise-01-domain-entities.md](#).

CREATING AND USING OBJECTS

Objects are created from classes and used to access data and call methods. (1 of 3)

? 1. WHAT DOES IT MEAN?

An object is an instance of a class. You create objects to represent real-world entities and use them to work with data and behavior.

RELATIONSHIP

Class (Car)
Blueprint

↓ *creates*

Object (myCar)
Real-world entity / Instance

⚙️ 2. HOW TO CREATE AN OBJECT IN JAVA?

SYNTAX

```
ClassName objectName = new ClassName();
```

NOTES

- new keyword allocates memory for the object
- Constructor is called to initialize the object
- objectName is a reference to the object

KEY TAKEAWAY An object is an instance of a class — the new keyword allocates memory and calls the constructor.

Creating and Using Objects

In Java, objects are created from classes and used to access variables and methods.

1. Create a Class (Blueprint)

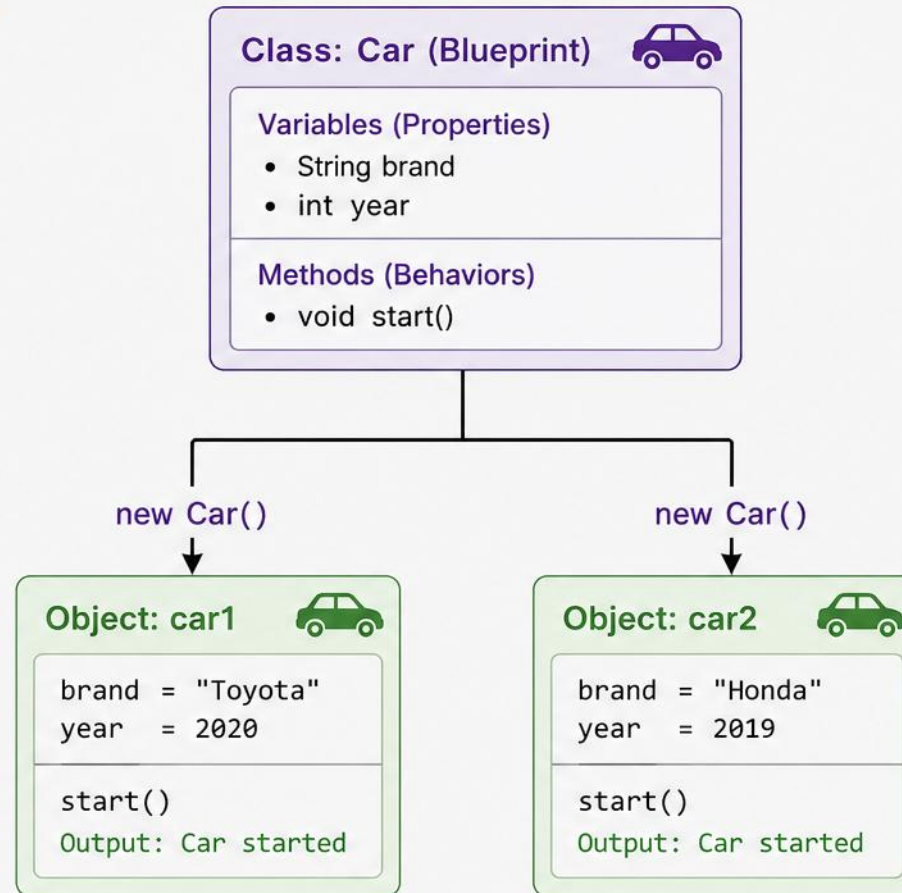
```
class Car {  
    String brand;  
    int year;  
  
    void start() {  
        System.out.println("Car started");  
    }  
}
```

2. Create Objects

```
Car car1 = new Car();    // object 1  
Car car2 = new Car();    // object 2
```

3. Use Objects

```
car1.brand = "Toyota";  
car1.year = 2020;  
car1.start();  
  
car2.brand = "Honda";  
car2.year = 2019;  
car2.start();
```



Key Points

- A class is a blueprint for creating objects.
- Use the `new` keyword to create objects.
- Each object has its own copy of variables.
- Objects communicate with methods.



CREATING AND USING OBJECTS

3. Example: Car Class and Objects, and the resulting output. (2 of 3)

```
class Car {  
    String brand; int speed;  
  
    void displayInfo() {  
        System.out.println("Brand: " + brand +  
            ", Speed: " + speed + " km/h");  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        Car car1 = new Car();  
        Car car2 = new Car();  
        car1.brand = "Toyota"; car1.speed = 120;  
        car2.brand = "Honda"; car2.speed = 80;  
        car1.displayInfo();  
        car2.displayInfo();  
    }  
}
```

5. OUTPUT

```
Brand: Toyota, Speed: 120 km/h  
Brand: Honda, Speed: 80 km/h
```

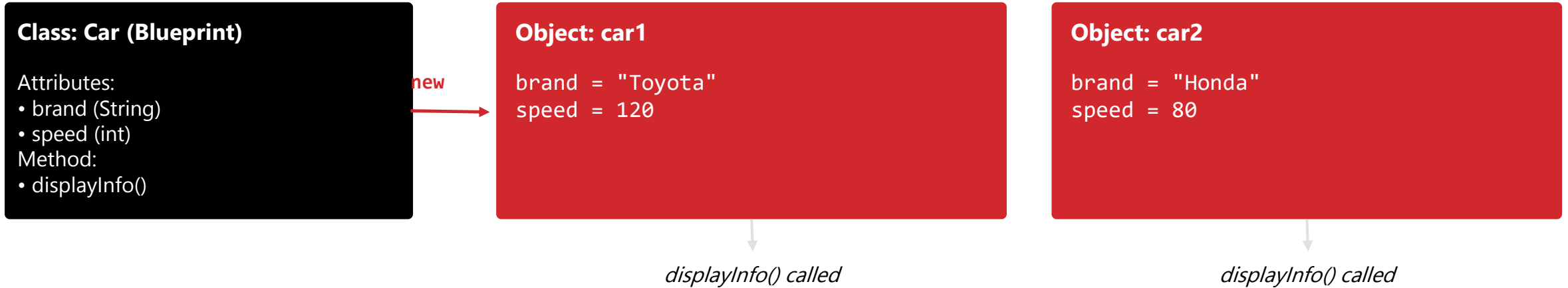
BEST PRACTICES

- Use meaningful names for classes and objects
- Initialize object state properly
- Encapsulate data (private fields, public methods)
- Keep objects focused on a single responsibility

KEY TAKEAWAY Two objects, one class — each car1 and car2 holds its own brand and speed.

CREATING AND USING OBJECTS

4. How It Works — from class definition to two independent objects. (3 of 3)



✓ KEY POINTS

- A class is a blueprint; an object is an instance of that class
- Use the `new` keyword to create objects
- Each object has its own copy of data (state stored separately)
- Methods define behavior; you call them using the object
- Multiple objects can be created from a single class

🏆 KEY TAKEAWAY

Creating and using objects is the foundation of working with real-world data in OOP. It helps build modular, reusable, and maintainable Java applications.

KEY TAKEAWAY Creating and using objects is the foundation of working with real-world data in OOP.

CONSTRUCTORS IN JAVA

Special methods that initialize objects, called automatically when created with new. (1 of 4)

? 1. WHAT IS A CONSTRUCTOR?

- Has the same name as the class
- Has no return type — not even void
- Called automatically when an object is created
- Used to initialize object state (variables)

ANALOGY

A constructor is like a builder of a house. It sets up and prepares the object when it is created.

⚙️ 2. SYNTAX

```
class ClassName {  
    // Constructor  
    ClassName() {  
        // initialization code  
    }  
}
```

KEY POINTS

- Constructor name = Class name
- No return type
- Called automatically by new

📁 3. TYPES OF CONSTRUCTORS

Default (No-Argument)

Takes no parameters. Provided by the compiler if none is defined.
new Car();

Parameterized

Takes one or more parameters to initialize with specific values.
new Car("Toyota", 120);

KEY TAKEAWAY Constructors ensure every object starts in a valid, consistent state.

Constructors in Java

A constructor is a special method used to initialize objects.
It is called automatically when an object is created.



Key Points

- The name of the constructor is the same as the class name.
- It has **no return type**, not even void.
- It is called automatically when an object is created using **new**.
- Used to initialize object state.
- Constructors can be overloaded.
- If no constructor is defined, Java provides a **default no-arg constructor**.

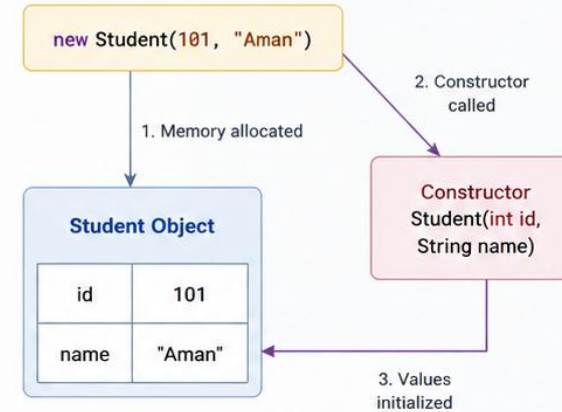


Example

```
class Student {  
    int id;  
    String name;  
  
    // Constructor  
    Student(int id, String name) {  
        this.id = id;  
        this.name = name;  
    }  
  
    public class Main {  
        public static void main(String[] args) {  
            Student s1 = new Student(101, "Aman");  
            System.out.println(s1.id + " " + s1.name);  
        }  
    }  
}
```



How It Works



Types of Constructors

1. Default (No-Arg) Constructor

Takes no parameters.
Provided by Java if no constructor is defined.

```
class Student {  
    int id;  
    String name;  
  
    // Default constructor  
    Student() {  
        id = 0;  
        name = "Unknown";  
    }  
}
```

Usage:
`Student s = new Student();`

2. Parameterized Constructor

Takes parameters to initialize
object with specific values.

```
class Student {  
    int id;  
    String name;  
  
    // Parameterized constructor  
    Student(int id, String name) {  
        this.id = id;  
        this.name = name;  
    }  
}
```

Usage:
`Student s = new Student(101, "Aman");`



Rules & Best Practices

- ✓ Constructor name must match the class name (exact case).
- ✓ Constructor has no return type.
- ✓ Use constructors to ensure objects are properly initialized.
- i Use constructor overloading for flexibility.
- ★ Keep constructors simple and focused on initialization.
- ⚠ Avoid heavy logic in constructors.



Summary

Constructors ensure that an object is in a valid state when it is created.
They play a vital role in initializing and maintaining object data.



CONSTRUCTORS IN JAVA

4. Example — the Car class with both constructors. 5. How Constructors Work. (2 of 4)

```
class Car {  
    String brand;  
    int speed;  
  
    // Default constructor  
    Car() {  
        brand = "Unknown";  
        speed = 0;  
    }  
  
    // Parameterized constructor  
    Car(String brand, int speed) {  
        this.brand = brand;  
        this.speed = speed;  
    }  
  
    void display() {  
        System.out.println("Brand: " + brand +  
            ", Speed: " + speed + " km/h");  
    }  
}
```

5. HOW CONSTRUCTORS WORK

- 1 Object created using new — memory is allocated
- 2 The appropriate constructor is called
- 3 Constructor initializes the member variables
- 4 The object is ready to use
- 5 Methods can be called on the object

MEMORY (HEAP) — Car Object

brand → "Toyota" speed → 120

KEY TAKEAWAY new allocates memory, the constructor runs, fields are initialized, then the object is ready to use.

CONSTRUCTORS IN JAVA

6. Example Output. 7. Constructor Overloading — multiple constructors, different parameter lists. (3 of 4)

```
Car car1 = new Car();  
Car car2 = new Car("Toyota", 120);  
  
car1.display();  
car2.display();
```

OUTPUT

```
Brand: Unknown, Speed: 0 km/h  
Brand: Toyota, Speed: 120 km/h
```

```
class Student {  
    String name;  
    int age;  
    double cgpa;  
  
    Student() {  
        name = "Unknown"; age = 0; cgpa = 0.0;  
    }  
    Student(String name) {  
        this.name = name;  
    }  
    Student(String name, int age) {  
        this.name = name; this.age = age;  
    }  
    Student(String name, int age, double cgpa) {  
        this.name = name; this.age = age;  
        this.cgpa = cgpa;  
    }  
}
```

KEY TAKEAWAY A class can have multiple constructors with different parameter lists — constructor overloading.

CONSTRUCTORS IN JAVA

8. Rules for Constructors. 9. this() — Calling Another Constructor. (4 of 4)

8. RULES FOR CONSTRUCTORS

- Constructor name must be the same as the class name
- Constructor must not have a return type
- Constructors are called automatically
- If no constructor is provided, Java creates a default constructor
- this() can be used to call another constructor in the same class

KEY TAKEAWAY

Constructors ensure objects start with a valid and consistent state. Constructor overloading provides flexibility.

9. THIS() — CALLING ANOTHER CONSTRUCTOR

```
class Car {  
    String brand; int speed;  
    Car() {  
        this("Unknown", 0);  
    }  
    Car(String brand, int speed) {  
        this.brand = brand; this.speed = speed;  
    }  
}
```

this() must be the first statement in the constructor body.

- Constructors initialize objects when they are created
- Use parameterized constructors to pass values while creating objects

KEY TAKEAWAY Constructors initialize objects; this() avoids duplicated initialization logic across overloaded constructors.

UNDERSTANDING ENCAPSULATION

Bundling data and methods into one unit, restricting direct access to some components. (1 of 3)

? 1. WHAT IS ENCAPSULATION?

Encapsulation hides the internal state of an object and allows access only through public methods (getters and setters).

Encapsulation provides:

- Protects data from unauthorized access
- Controls how data is modified and read
- Improves maintainability and flexibility

⚙️ 2. HOW ENCAPSULATION WORKS

🔒 Private Data (Hidden)

accountNumber, balance — these fields cannot be accessed directly from the outside world.



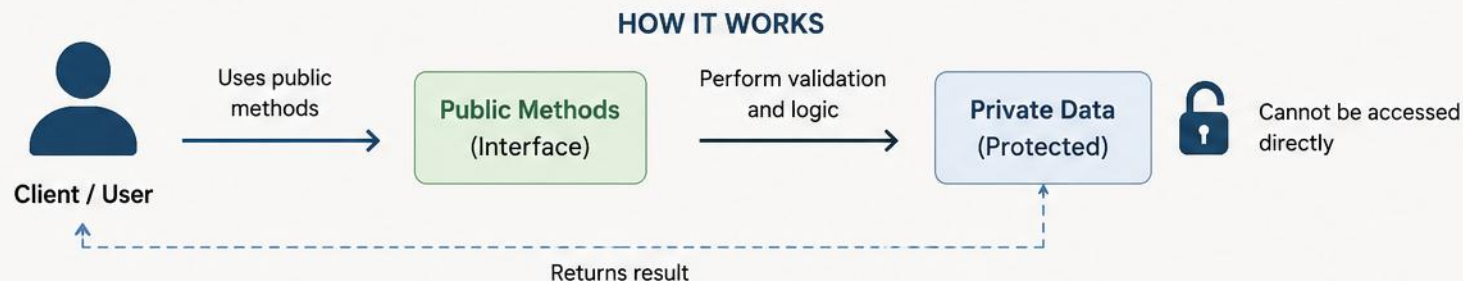
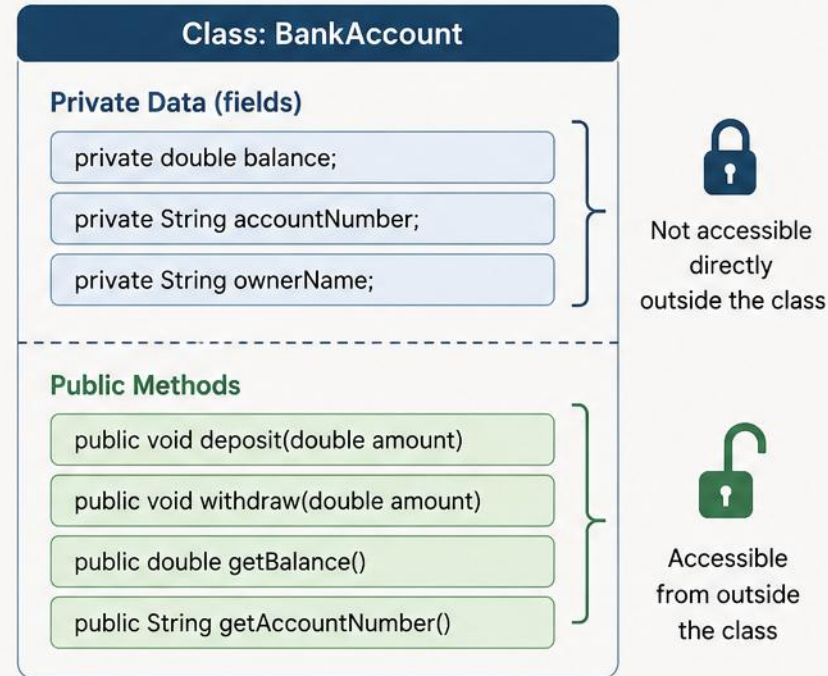
🔓 Public Methods (Access via methods)

getBalance(), deposit(double amount), withdraw(double amount)
Access to private data is provided only through these methods.

KEY TAKEAWAY Private fields, public methods: outside code can only touch what the class chooses to expose.

ENCAPSULATION IN JAVA

Encapsulation is the OOP principle of wrapping data (variables) and code (methods) together as a single unit and restricting direct access to some of an object's components.



UNDERSTANDING ENCAPSULATION

3. Real-World Analogy. 4. Example in Java — the BankAccount class. (2 of 3)



3. REAL-WORLD ANALOGY — BANK EXAMPLE



You (Customer)



Bank Vault (Private Data)

You cannot access the vault directly. Instead, you interact through bank staff (Public Methods) to:

- Deposit money
- Withdraw money
- Check account balance

```
public class BankAccount {  
    private String accountNumber;  
    private double balance;  
  
    public double getBalance() { return balance; }  
  
    public void deposit(double amount) {  
        if (amount > 0) { balance += amount; }  
    }  
  
    public void withdraw(double amount) {  
        if (amount > 0 && amount <= balance) {  
            balance -= amount;  
        }  
    }  
}
```

KEY TAKEAWAY You interact with a bank vault through staff, never directly — the same idea protects an object's data.

UNDERSTANDING ENCAPSULATION

What we can and cannot do, the benefits, and the key takeaways. (3 of 3)

```
// ❌ What We CANNOT Do
BankAccount acc = new BankAccount();
// acc.balance = 1000;           // Error!
// acc.accountNumber = "12345"; // Error!
```

```
// ✅ What We CAN Do
BankAccount acc = new BankAccount();
acc.deposit(1000);
System.out.println("Balance: " + acc.getBalance());
acc.withdraw(200);
System.out.println("Balance: " + acc.getBalance());
```

★ 5. BENEFITS OF ENCAPSULATION

Data Protection

Prevents direct access to sensitive data and maintains data integrity.

Controlled Access

Allows validation and rules before data is modified.

Maintainability

Changes in internal implementation do not affect code that uses the class.

Flexibility

Internal details can be changed without impacting the outside world.

🏆 6. KEY TAKEAWAYS

- Encapsulation = Data Hiding + Controlled Access
- Use private for fields; provide public getters/setters
- A fundamental principle of OOP for building robust applications

KEY TAKEAWAY Remember: encapsulation isn't about making everything private — it's about hiding what shouldn't be seen. Result: more secure, reliable, easy-to-maintain code.

ACCESS MODIFIERS IN JAVA

They determine where in the program your code can access a particular member. (1 of 3)

 **private**

Accessible only within the same class.

 **default (no modifier)**

Accessible within the same package.

 **protected**

Accessible within the same package and by subclasses (including subclasses in other packages).

 **public**

Accessible from anywhere in the application.

2. ACCESS LEVELS — AT A GLANCE

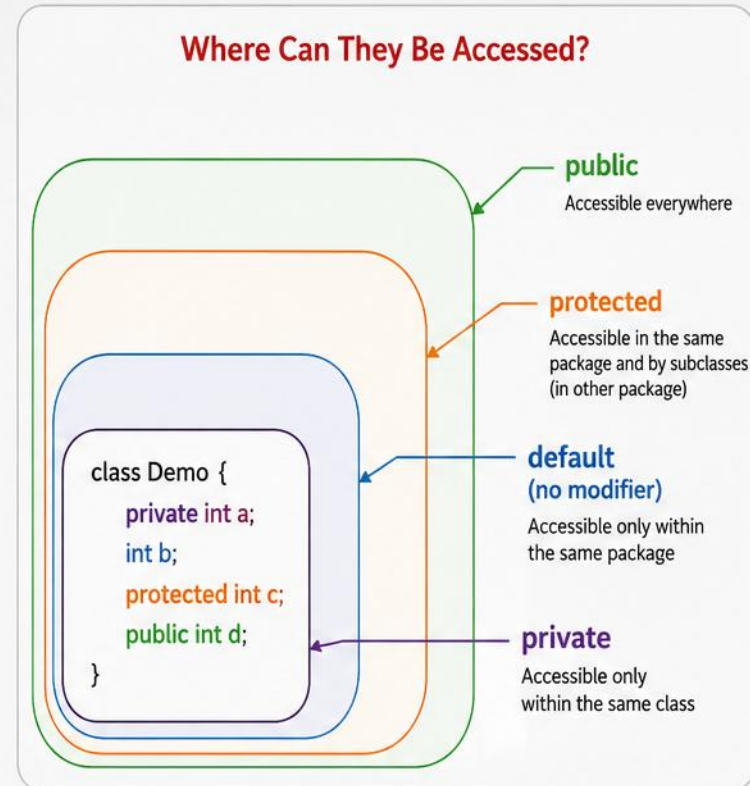
MODIFIER	SAME CLASS	SAME PACKAGE	SUBCLASS (OTHER PKG)	OUTSIDE PACKAGE
private	✓	✗	✗	✗
default	✓	✓	✗	✗
protected	✓	✓	✓	✗
public	✓	✓	✓	✓

KEY TAKEAWAY More restrictive = safer. Public is the most permissive; private is the most restrictive.

Access Modifiers in Java

Access modifiers control the visibility (scope) of classes, methods, and variables.

Modifier	Within Same Class	Within Same Package	Subclasses in Different Package	Outside Package
private 	✓	✗	✗	✗
default (no modifier) 	✓	✓	✗	✗
protected 	✓	✓	✓	✗
public 	✓	✓	✓	✓



private

- Accessible only within the same class.
- Not visible outside the class.



default
(no modifier)

- Accessible within the same package.
- Not visible in other packages.



protected

- Accessible within the same package.
- Also accessible in subclasses even if they are in a different package.



public

- Accessible from anywhere.
- No restrictions.



ACCESS MODIFIERS

ACCESS MODIFIERS IN JAVA

3. Key Points. 4. Example — all four modifiers used in one class. (2 of 3)

📌 3. KEY POINTS

- Top-level classes can be public or default (no modifier) only
- Member variables, methods, and constructors can use any of the four access modifiers
- If no modifier is specified, it is default by default
- More restrictive access is generally safer and encourages encapsulation
- Public is the most permissive; private is the most restrictive

```
package com.bank.account;

public class Account {
    private String accountNumber; // private
    double balance;               // default
    protected double interestRate; // protected
    public String bankName;       // public

    // Constructor
    public Account(String accountNumber,
                    double balance) {
        this.accountNumber = accountNumber;
        this.balance = balance;
        this.interestRate = 4.5;
        this.bankName = "Trusted Bank";
    }
}
```

KEY TAKEAWAY One class, four modifiers — each field's visibility is chosen deliberately.

ACCESS MODIFIERS IN JAVA

5. Example Scenario. 6. Real-World Analogy. (3 of 3)

```
// SavingsAccount is in com.bank.savings
// It extends Account, in com.bank.account
package com.bank.savings;
import com.bank.account.Account;

public class SavingsAccount extends Account {
    public void showInfo() {
        // accountNumber -> Not accessible (private)
        // balance -> Not accessible (default)
        // interestRate -> Accessible (protected)
        System.out.println(interestRate); // OK
        // bankName -> Accessible (public)
        System.out.println(bankName);      // OK
        // System.out.println(accountNumber); // Error
    }
}
```



6. REAL-WORLD ANALOGY



private

Manager's safe — only the manager can access it.



default

Office rooms inside the bank — staff in the same office can access.



protected

Departments and authorized branches — accessible to related departments.



public

Public information — available to customers and everyone.



BEST PRACTICE Use the most restrictive access level that still allows the class to function correctly.

KEY TAKEAWAY Access modifiers are fundamental to OOP — they enforce encapsulation, protect data, and build secure, maintainable applications.

GETTERS AND SETTERS IN JAVA

Public methods that access and update private fields, supporting encapsulation. (1 of 3)



Encapsulation

Keeps data private and protects it from direct access.



Validation

Allows validation before assigning values.



Flexibility

Internal representation can change without affecting other classes.



Read-Only / Write-Control

Makes fields read-only (getter only) or write-only (setter with rules).



2. HOW THEY WORK

Other Class (Client) uses public methods to interact with your class →

Private Fields **name, balance**

*Read data via getter
Update data via setter*

Public Methods

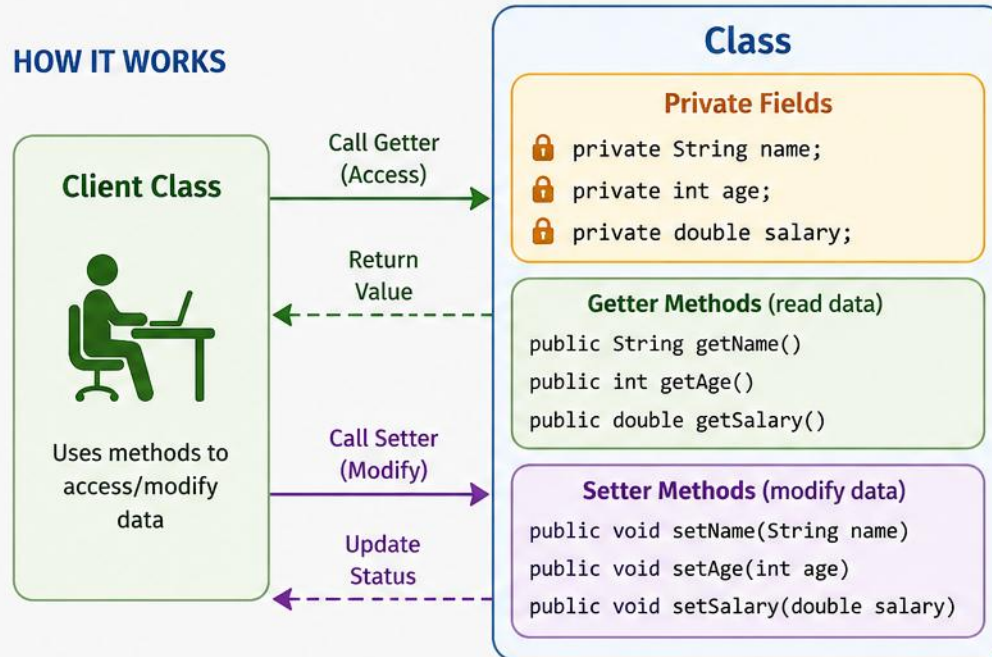
```
getName() setName(String name)  
getBalance() setBalance(double balance)
```

KEY TAKEAWAY Getters and setters let outside code interact with your class only through controlled, validated methods.

GETTERS AND SETTERS IN JAVA

Getters and setters are methods used to access and modify the private fields (attributes) of a class. They provide controlled access and help in encapsulation.

HOW IT WORKS



BENEFITS



Encapsulation
Protects data by controlling access.



Data Validation
Add checks in setters before updating.



Flexibility
Change internal implementation without affecting client code.



Read-Only Access
Provide only getter if data should not be modified.

EXAMPLE

```
public class Employee {  
    private String name;  
    private int age;  
    private double salary;  
  
    // Getter methods  
    public String getName() { return name; }  
    public int getAge() { return age; }  
    public double getSalary() { return salary; }  
  
    // Setter methods  
    public void setName(String name) { this.name = name; }  
    public void setAge(int age) { this.age = age; }  
    public void setSalary(double salary) { this.salary = salary; }  
}
```

BEST PRACTICES

- ✓ Keep fields private.
- ✓ Use meaningful method names: getX(), setX().
- ✓ Validate data in setter methods.
- ✓ Avoid placing business logic in getters and setters.



GETTERS AND SETTERS IN JAVA

3. Example: BankAccount Class — getters and setters with validation. (2 of 3)

```
public class BankAccount {  
    private String accountNumber;    private double balance;  
    public BankAccount(String accountNumber, double balance) {  
        this.accountNumber = accountNumber;    this.balance = balance;  
    }  
    public String getAccountNumber() { return accountNumber; }  
    public void setAccountNumber(String accountNumber) {  
        if (accountNumber != null && !accountNumber.isBlank()) {  
            this.accountNumber = accountNumber;  
        } else { System.out.println("Invalid account number!"); }  
    }  
    public double getBalance() { return balance; }  
    public void setBalance(double balance) {  
        if (balance >= 0) { this.balance = balance; }  
        else { System.out.println("Balance cannot be negative!"); }  
    }  
}
```

KEY TAKEAWAY Setters validate before assigning — an empty account number or negative balance is rejected.

GETTERS AND SETTERS IN JAVA

4. Using Getters and Setters. 5. Rules. 6. Variations. (3 of 3)

```
BankAccount account =
    new BankAccount("123456", 1000.0);

// Using getters
System.out.println("Account: " +
    account.getAccountNumber());
System.out.println("Balance: " + account.getBalance());

// Using setters
account.setBalance(1500.0);
account.setAccountNumber("987654");
```

 OUTPUT

```
Account: 123456 | Balance: 1000.0
Updated Account: 987654 | Updated Balance: 1500.0
```

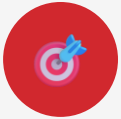
RULE	DESCRIPTION
Naming	getFieldName() / setFieldName()
Getter return	Same as the field type
Setter return	Usually void
Access modifier	Always public
Validation	Add checks in setters
Read-Only	getName() – no setter
Write-Only	setPassword() – no getter
Computed	getAnnualInterest() = balance*rate*12

Remember: never expose private fields directly. Goal: protect the data, control the access, keep the code flexible.

KEY TAKEAWAY Getters and setters are a core part of encapsulation — always validate data in setter methods.

TRY IT YOURSELF — EXERCISE 2: ENCAPSULATION PRACTICE

Reinforces: private fields, access modifiers, and getters/setters you just saw.



1

Goal

Create Account with a private balance and deposit / withdraw methods



2

File

Account.java in
examples/module-03-
exercises/



3

Key concept

private double balance
— no public field writes,
all changes go through
methods



4

Compile & Run

```
javac Account.java  
EncapsulationDemo.java  
&& java  
EncapsulationDemo
```



5

Reference

exercises/exercise-02-
encapsulation.md

TRY IT NOW Complete Exercise 2 before continuing — see [exercises/exercise-02-encapsulation.md](#).

UNDERSTANDING INHERITANCE

A new class acquires the properties and behaviors of an existing class — an "is-a" relationship. (1 of 4)

1. KEY CONCEPTS

- The class whose members are inherited is the Superclass (Parent class)
- The class that inherits is the Subclass (Child class)
- Subclasses can reuse the fields and methods of the superclass
- Subclasses can add their own fields and methods
- Promotes reusability, extensibility, and maintainability

REAL-WORLD ANALOGY

A Car is a Vehicle (inherits properties like wheels and engine). A Car also has its own features: music system, sunroof, and other car-specific features.

2. HOW INHERITANCE WORKS

Superclass (Parent): Vehicle

Fields: brand, maxSpeed
Methods: start(), stop()

inherited by
↓

Subclass (Child): Car

Additional: numDoors, playMusic()

Relationship: Car IS-A Vehicle

KEY TAKEAWAY Use extends to inherit. Design inheritance only where a clear IS-A relationship exists.

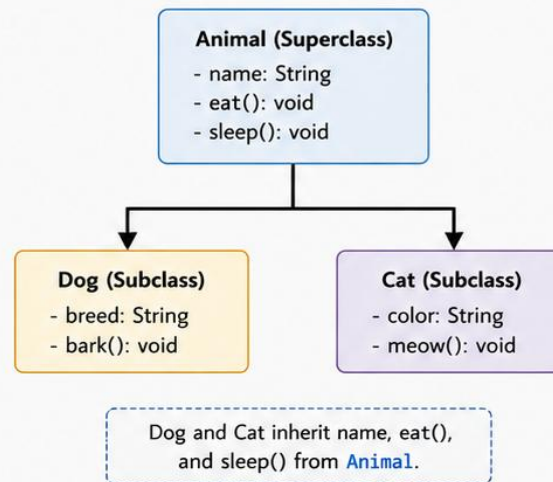
INHERITANCE IN JAVA

Inheritance is a mechanism in Java where one class acquires the properties (fields and methods) of another class.

KEY POINTS

- Inheritance promotes code reusability and establishes a relationship between classes.
- The class whose members are inherited is called the Superclass (Parent).
- The class that inherits is called the Subclass (Child).
- In Java, a class can extend only one class (Single Inheritance).
- Use the `extends` keyword to inherit a class.

CLASS HIERARCHY (EXAMPLE)

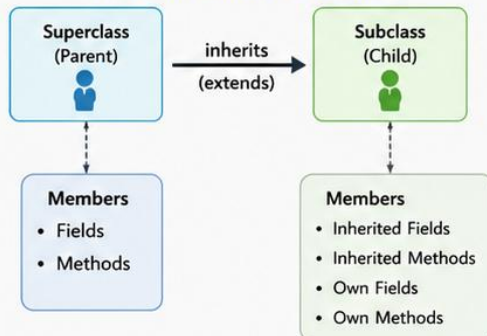


EXAMPLE CODE

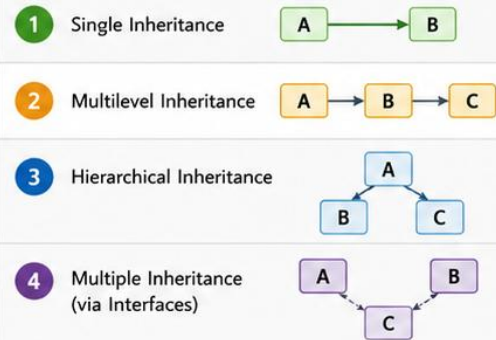
```
// Superclass
class Animal {
    String name;
    void eat() {
        System.out.println("Eating...");
    }
    void sleep() {
        System.out.println("Sleeping...");
    }
}

// Subclass
class Dog extends Animal {
    String breed;
    void bark() {
        System.out.println("Barking...");
    }
}
```

HOW INHERITANCE WORKS



TYPE OF INHERITANCE IN JAVA



BENEFITS



Code Reusability

Reusing fields and methods of existing classes.



Extensibility

Allows adding new features in the subclass.



Maintainability

Improves code organization and readability.



Polymorphism Support

Enables method overriding and dynamic binding.



Inheritance represents an IS-A relationship.
Example: Dog IS-A Animal, Cat IS-A Animal.

Use `extends` keyword to create a subclass.

```
class SubClass extends SuperClass { ... }
```



UNDERSTANDING INHERITANCE

3. Syntax in Java. 4. Example — using an inherited Car object. (2 of 4)

```
// Superclass
class Vehicle {
    String brand;
    int maxSpeed;

    void start() {
        System.out.println("Vehicle started");
    }
    void stop() {
        System.out.println("Vehicle stopped");
    }
}

// Subclass
class Car extends Vehicle {
    int numDoors;
    void playMusic() {
        System.out.println("Playing music");
    }
}
```

```
public class TestInheritance {
    public static void main(String[] args) {
        Car myCar = new Car();
        myCar.start(); myCar.stop();           // inherited
        myCar.numDoors = 4;                     // own
        myCar.playMusic();
        myCar.brand = "Toyota";                 // inherited
        myCar.maxSpeed = 180;
        System.out.println("Brand: " + myCar.brand);
        System.out.println("Max Speed: " + myCar.maxSpeed);
    }
}
```

OUTPUT: Vehicle started → Vehicle stopped → Playing music
→ Brand: Toyota → Max Speed: 180

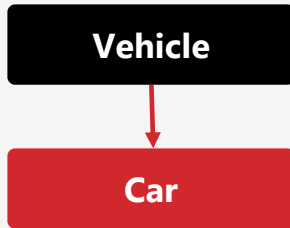
KEY TAKEAWAY myCar can use both its own members and everything it inherited from Vehicle.

UNDERSTANDING INHERITANCE

5. Types of Inheritance in Java — three structural forms. (3 of 4)

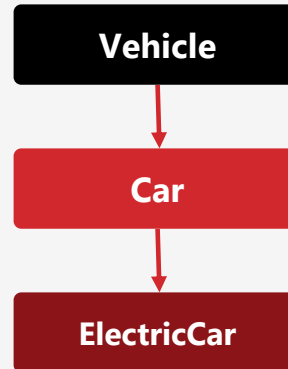
SINGLE

A subclass inherits from exactly one superclass.



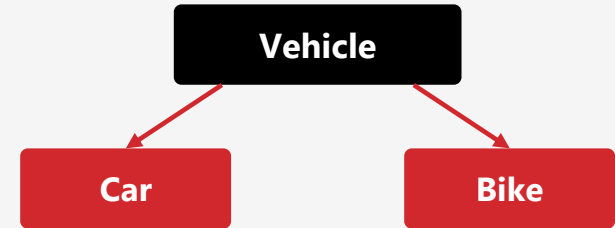
MULTILEVEL

A subclass inherits from a class that is itself a subclass.



HIERARCHICAL

Multiple subclasses inherit from the same superclass.



💡 NOTE — METHOD OVERRIDING

A subclass can provide its own implementation of a method already defined in its superclass — covered in detail on the next slide.

KEY TAKEAWAY Java supports single, multilevel, and hierarchical class inheritance — but not multiple class inheritance.

UNDERSTANDING INHERITANCE

6. Method Overriding. 7. super Keyword. 8. Key Takeaways. (4 of 4)

```
class Vehicle {  
    void start() {  
        System.out.println("Vehicle started");  
    }  
}  
  
class Car extends Vehicle {  
    @Override  
    void start() {  
        System.out.println("Car started with key");  
    }  
}
```

6. METHOD OVERRIDING — NOTE

The overridden method must have the same method name, the same parameter list, and the same (or compatible) return type.

```
class Vehicle {  
    String brand;  
    Vehicle(String brand) { this.brand = brand; }  
}  
  
class Car extends Vehicle {  
    int numDoors;  
    Car(String brand, int numDoors) {  
        super(brand); // calls parent ctor  
        this.numDoors = numDoors;  
    }  
}
```

7. **super KEYWORD** — accesses superclass members and calls the superclass constructor.

8. KEY TAKEAWAYS

- Inheritance models the IS-A relationship; use extends to inherit
- A subclass gets all accessible members of the superclass
- Method overriding allows specialized implementations
- Use super to access parent members and constructors

Goal: use inheritance wisely to model real-world relationships and write better OOP Java programs

KEY TAKEAWAY Good Practice: design inheritance only when there's a clear IS-A relationship. Remember: it leads to clean, reusable, extensible code.

SINGLE INHERITANCE IN JAVA

A subclass inherits from only one superclass (parent class). (1 of 5)

? 1. WHAT IS SINGLE INHERITANCE?

A subclass inherits the fields and methods of only one superclass.

Superclass (Parent): Animal
Fields: name, age
Method: eat()

↓ extends

Subclass (Child): Dog
Field: breed
Method: bark()

Note: A Dog is an Animal. It can use all properties and methods of Animal.

⚙️ 2. SYNTAX

```
class SuperClass {  
    // fields and methods  
}  
  
class SubClass extends SuperClass {  
    // additional fields and methods  
}
```

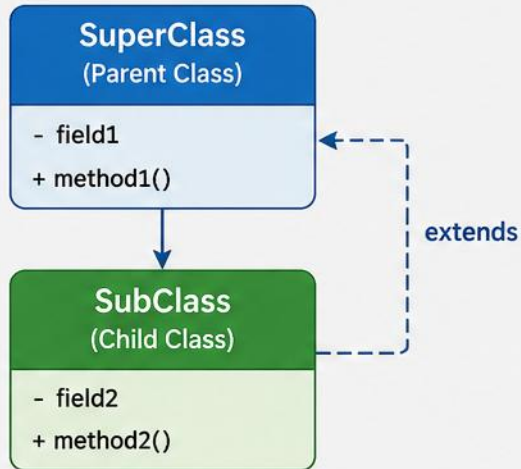
- SubClass extends SuperClass
- SubClass inherits all accessible members of SuperClass
- Use the extends keyword

KEY TAKEAWAY A Dog is an Animal — it inherits Animal's members and adds its own.

SINGLE INHERITANCE IN JAVA

Single inheritance is a feature in Java where a subclass inherits properties and behaviors from a single superclass.

CLASS DIAGRAM

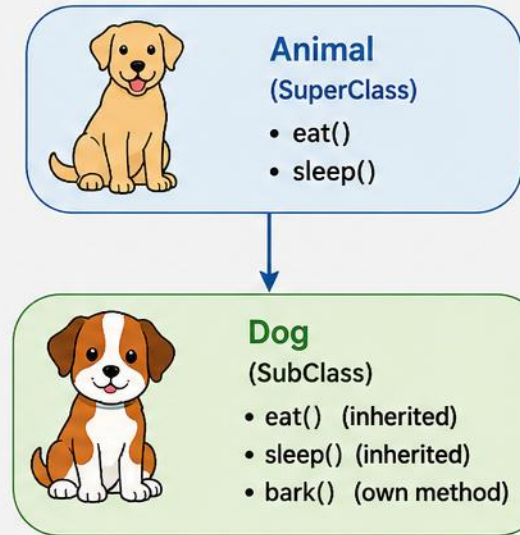


CODE EXAMPLE

```
// Parent Class
class Animal {
    void eat() {
        System.out.println("Animal is eating");
    }
}

// Child Class
class Dog extends Animal {
    void bark() {
        System.out.println("Dog is barking");
    }
}
```

VISUAL REPRESENTATION



KEY POINTS

- ✓ A subclass inherits fields and methods from one superclass only.
- ✓ Use the `extends` keyword to achieve single inheritance.
- ✓ The subclass can add its own fields and methods.
- ✓ Promotes code reusability and establishes an **IS-A** relationship.



SINGLE INHERITANCE IN JAVA

3. Example — the Animal superclass and Dog subclass. (2 of 5)

```
// Superclass: Animal.java
class Animal {

    String name;
    int age;

    void eat() {
        System.out.println(name + " is eating");
    }

    void sleep() {
        System.out.println(name + " is sleeping");
    }
}
```

```
// Subclass: Dog.java
class Dog extends Animal {

    String breed;

    void bark() {
        System.out.println(name + " is barking");
    }
}
```

KEY TAKEAWAY Dog reuses Animal's name, age, eat(), and sleep() — and adds breed and bark() of its own.

SINGLE INHERITANCE IN JAVA

4. Usage — *TestInheritance.java* puts both classes to work. (3 of 5)

```
public class TestInheritance {  
    public static void main(String[] args) {  
  
        Dog d = new Dog();  
  
        // Inherited fields  
        d.name = "Tommy";  
        d.age = 3;  
  
        // Own field  
        d.breed = "Labrador";  
  
        // Inherited methods  
        d.eat();      // from Animal  
        d.sleep();    // from Animal  
  
        // Own method  
        d.bark();     // from Dog  
    }  
}
```

OUTPUT

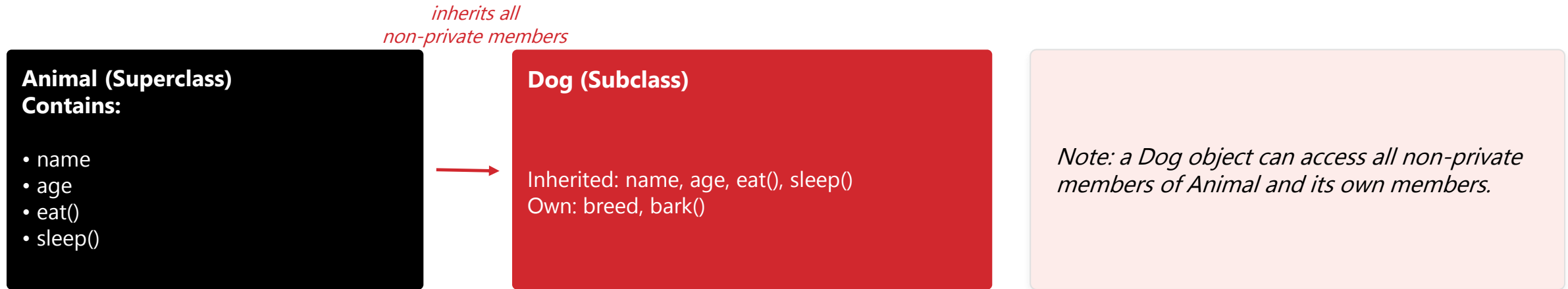
```
Tommy is eating  
Tommy is sleeping  
Tommy is barking
```

One Dog object reaches all three methods — two inherited from Animal, one its own.

KEY TAKEAWAY Inherited fields and methods behave exactly like the subclass's own — no extra syntax needed to use them.

SINGLE INHERITANCE IN JAVA

5. How It Works. 6. Key Points. (4 of 5)



✓ 6. KEY POINTS

- A subclass can inherit from only one superclass.
- A subclass gets all non-private fields and methods of its superclass.
- Private members of the superclass are not inherited.
- The constructor of the superclass is called first.
- A subclass can have its own fields and methods in addition to inherited ones.
- Use the super keyword to access superclass members (overridden or hidden).

KEY TAKEAWAY A subclass inherits everything non-private from its one superclass, then adds its own members.

SINGLE INHERITANCE IN JAVA

7. *super* Keyword Example. 8. Summary. (5 of 5)

```
class Animal {
    String name = "Animal";
    Animal() {
        System.out.println("Animal constructor");
    }
    void show() {
        System.out.println("Animal show()");
    }
}

class Dog extends Animal {
    String name = "Dog";
    Dog() {
        super(); // calls Animal constructor
        System.out.println("Dog constructor");
    }
    void show() {
        super.show(); // calls Animal show()
        System.out.println("Dog show()");
    }
}
```

8. SUMMARY

- Models a real-world IS-A relationship (e.g., Dog IS-A Animal)
- Promotes code reusability and reduces duplication
- Helps build scalable and maintainable applications
- Forms the foundation for multilevel and hierarchical inheritance

REMEMBER

Single inheritance means "one parent, many children." Java does not support multiple inheritance with classes, but it does support it through interfaces.

KEY TAKEAWAY Single inheritance models real-world IS-A relationships and reduces code duplication.

METHOD OVERRIDING IN JAVA

A subclass provides a specific implementation of a method already defined in its superclass.

RULES FOR OVERRIDING

- Same method name and parameters
- Same or covariant return type
- Access level cannot be more restrictive
- static, final, private methods cannot be overridden
- Use @Override to catch mistakes

```
class Animal {  
    public void sound() {  
        System.out.println("...");  
    }  
}  
  
class Dog extends Animal {  
    @Override  
    public void sound() {  
        System.out.println("Bark");  
    }  
}
```

WHAT CANNOT BE OVERRIDDEN?

- final methods
- static methods (hidden, not overridden)
- private methods
- constructors

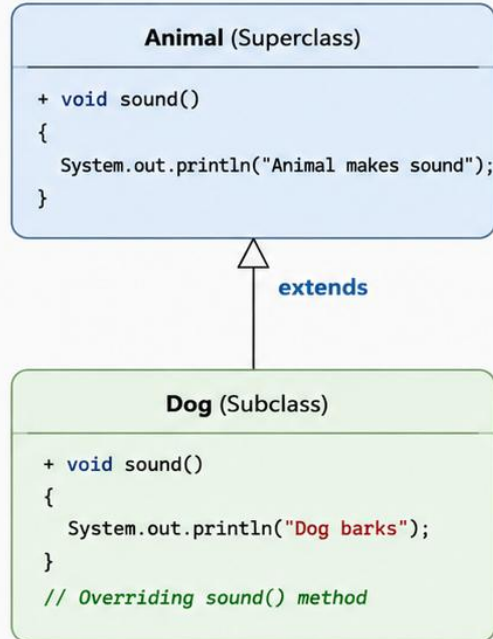
Overriding vs Overloading: overriding requires inheritance and runtime binding; overloading is same-class, compile-time binding.

KEY TAKEAWAY Method overriding is the key that unlocks runtime polymorphism in Java.

Method Overriding in Java

Method overriding allows a subclass to provide a specific implementation of a method that is already defined in its superclass.

Class Hierarchy

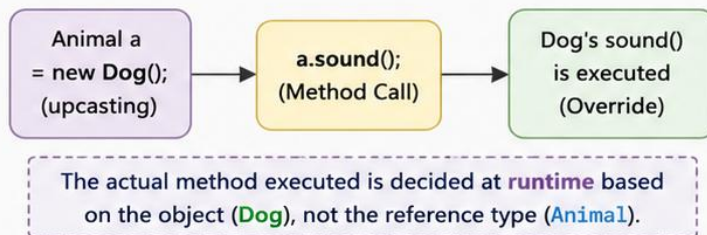


Code Example

```
1  class Animal {
2      void sound() {
3          System.out.println("Animal makes sound");
4      }
5  }
6
7  class Dog extends Animal {
8      @Override
9      void sound() {
10         System.out.println("Dog barks");
11     }
12 }
13
14 public class Test {
15     public static void main(String[] args) {
16         Animal a = new Dog(); // upcasting
17         a.sound();           // calls Dog's sound()
18     }
19 }
```

Subclass provides
its own
implementation

How It Works (Runtime)



Key Points

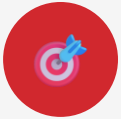
- ✓ Method name, parameters and return type must be same.
- ✓ Access level in subclass must be same or broader (cannot be more restrictive).
- ✓ @Override annotation is optional but recommended.
- ✓ Method overriding supports **runtime** polymorphism.
- ✓ It is used to provide specific behavior in the subclass.



Benefit: Achieves runtime polymorphism and promotes code reusability and extensibility.

TRY IT YOURSELF — EXERCISE 3: INHERITANCE SKETCH

Reinforces: extends and method overriding, as just covered.



1

Goal

Create SavingsAccount and CurrentAccount extending Account



2

File

SavingsAccount.java,
CurrentAccount.java in
examples/module-03-
exercises/



3

Key concept

extends Account, then
override withdraw() or
getAccountType()



4

Compile & Run

javac *.java && java
InheritanceDemo — call
both types via an
Account reference



5

Reference

exercises/exercise-03-
inheritance.md

TRY IT NOW Complete Exercise 3 before continuing — see [exercises/exercise-03-inheritance.md](#).

COMPILE-TIME POLYMORPHISM IN JAVA

Achieved by method overloading; the compiler decides which method runs.

✅ REQUIREMENTS FOR OVERLOADING

- Same method name, different parameter list (number, type, or order)
- Different return type ALONE is not enough
- Different access modifier alone is not enough

```
class Calculator {  
    int add(int a, int b) { return a+b; }  
    double add(double a, double b)  
        { return a+b; }  
    int add(int a, int b, int c)  
        { return a+b+c; }  
}
```

⚙️ HOW IT WORKS

```
Calculator c = new Calculator();  
c.add(10, 20);           → int add(int,int)  
c.add(10.5, 20.5);       → double add(double,double)  
c.add(10, 20, 30);       → int add(int,int,int)
```

★ BENEFITS

- Improves readability using one method name
- Provides flexibility to handle different inputs
- Also called static / early binding

KEY TAKEAWAY Method overloading is resolved by the compiler based on the method signature, not the return type.

COMPILE-TIME POLYMORPHISM

(Method Overloading in Java)



Compile-time polymorphism is achieved by method overloading – same method name with different parameter lists (number, type or order of parameters).

HOW IT WORKS



1. Multiple methods with the same name but different parameters



2. Compiler checks the method signature at compile time



3. Appropriate method is selected based on arguments passed



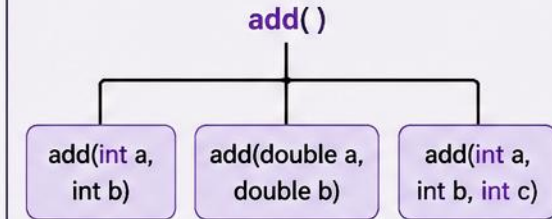
4. Method is bound at compile time

EXAMPLE

```
class Calculator {  
    int add(int a, int b) {  
        return a + b;  
    }  
  
    double add(double a, double b) {  
        return a + b;  
    }  
  
    int add(int a, int b, int c) {  
        return a + b + c;  
    }  
}  
  
public class Demo {  
    public static void main(String[] args) {  
        Calculator c = new Calculator();  
  
        System.out.println(c.add(10, 20));    // calls add(int, int)  
        System.out.println(c.add(10.5, 20.5)); // calls add(double, double)  
        System.out.println(c.add(10, 20, 30)); // calls add(int, int, int)  
    }  
}
```

METHOD OVERLOADING

Same Method Name



KEY POINTS

- Occurs within the same class.
- Method signature must be different.
- Return type can be same or different (not part of signature).
- Achieved at compile time.
- Improves code readability and reusability.

OUTPUT



```
30  
31.0  
60
```

RUNTIME POLYMORPHISM IN JAVA

Achieved by method overriding; the JVM decides which method runs.

🔑 KEY CONCEPTS

- Achieved using method overriding, in an IS-A relationship
- Method call is resolved at runtime, not compile time
- Reference type can be superclass; object type can be subclass

```
class Animal { void sound(){...} }  
class Dog extends Animal {  
    @Override  
    void sound() { ...; }  
}  
Animal a = new Dog();  
a.sound(); // Dog barks
```

💡 REAL-WORLD ANALOGY

A person (Animal reference) can behave differently (sound) based on their actual identity at runtime (Dog, Cat, Cow).

★ BENEFITS

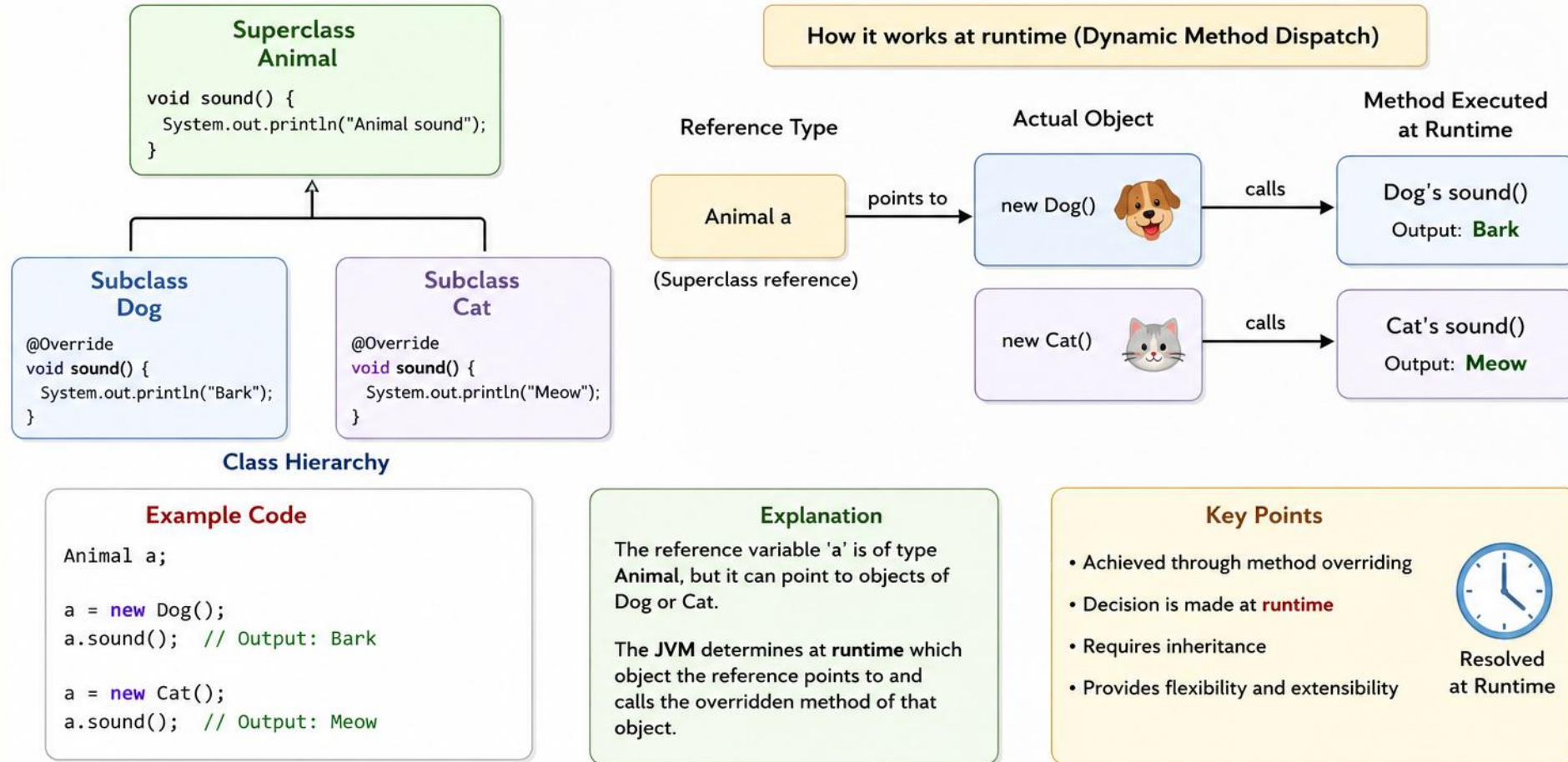
- Provides flexibility in code
- Allows a general interface with specific implementations
- Easier to extend and maintain large applications
- Supports the principle of loose coupling

KEY TAKEAWAY The JVM decides which overridden method to call based on the actual object, at runtime.

RUNTIME POLYMORPHISM IN JAVA

— Achieved through Method Overriding —

Runtime polymorphism allows a method call to be resolved at runtime.
The same method call behaves differently for different objects.



Runtime Polymorphism = One interface, multiple implementations, **decision at runtime**.

COMPILE-TIME VS RUNTIME POLYMORPHISM

Java achieves polymorphism two ways: overloading and overriding.

ASPECT	COMPILE-TIME (Overloading)	RUNTIME (Overriding)
Binding	Early binding	Late binding
Occurs in	Same class	Parent & child classes
Signature	Must differ	Must be identical
Decided by	Compiler	JVM at runtime
Performance	Faster	Slightly slower

Overloading: `int add(int,int)` vs `double add(double,double)`

Overriding: `Animal a = new Dog(); a.sound();` runs `Dog.sound()`

KEY TAKEAWAY Overloading is resolved by the compiler on the method signature; overriding is resolved by the JVM on the actual object.

DYNAMIC METHOD DISPATCH IN JAVA

How an overridden method call is resolved at runtime, based on the actual object type.



MEMORY VIEW

Stack (reference)	Heap (object)
a → 0x101	Dog object
	{ sound() →
Dog.sound() }	

Reference 'a' is typed Animal, but points to a Dog object — so Dog.sound() runs.

? WHY IT'S CALLED DYNAMIC

- The method call isn't decided until runtime
- The JVM inspects the actual object type
- Requires inheritance and method overriding
- Private, static, final methods are not dispatched dynamically

KEY TAKEAWAY Dynamic dispatch is runtime polymorphism, powered by inheritance and method overriding.

Dynamic Method Dispatch in Java

The JVM determines which overridden method to call at runtime based on the actual object type, not the reference type.

```
class Animal
{
    void sound() {
        System.out.println("Animal makes sound");
    }
}
```

extends

```
class Dog extends Animal
{
    @Override
    void sound() {
        System.out.println("Dog barks");
    }
}
```

extends

```
class Cat extends Animal
{
    @Override
    void sound() {
        System.out.println("Cat meows");
    }
}
```

```
public class Demo {
    public static void main(String[] args) {
        Animal a1 = new Dog();
        a1.sound();           // Dog barks

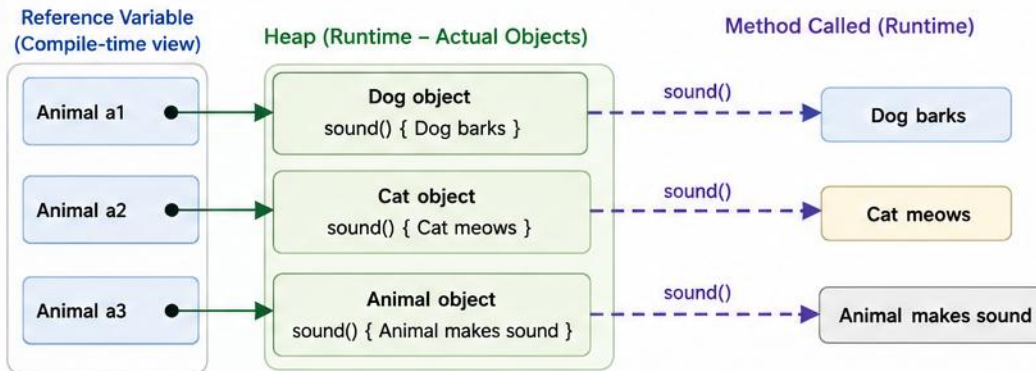
        Animal a2 = new Cat();
        a2.sound();           // Cat meows

        Animal a3 = new Animal();
        a3.sound();           // Animal makes sound
    }
}
```

What happens at runtime?

- 1 The reference type (Animal) has a reference to an object.
- 2 When sound() is called, the JVM looks at the actual object type.
- 3 The JVM calls the implementation of sound() in that actual class.

This process is called
Dynamic Method Dispatch
(Runtime Polymorphism)



Dynamic Method Dispatch enables flexibility and extensibility in Java by allowing subclass methods to override superclass methods and be invoked at runtime based on the actual object.



Key Takeaway



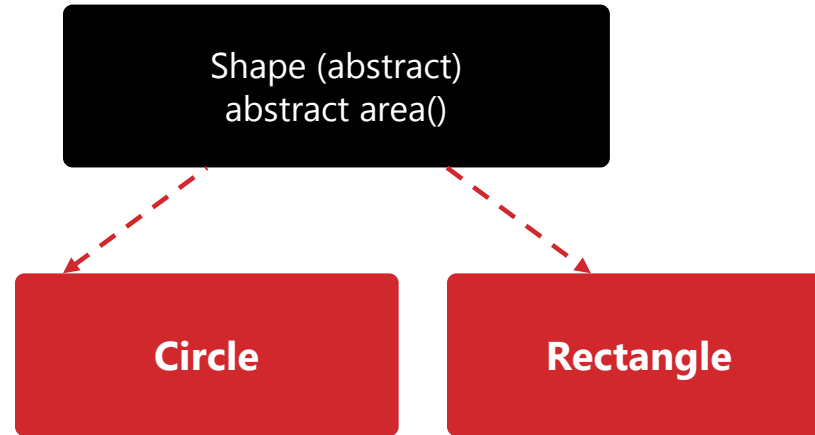
Which method gets executed is decided at runtime, based on the actual object referred to by the reference variable.

ABSTRACT CLASSES IN JAVA

A class that cannot be instantiated, designed to be extended. (1 of 2)

🔑 KEY FEATURES

- Cannot create objects of an abstract class
- Can have abstract methods (no body)
- Can have concrete methods (with body)
- Subclasses must implement all abstract methods



```
Shape s1 = new Circle("Red", 5.0);  
System.out.println(s1.area()); // 78.5
```

Both subclasses provide their own implementation of area().

! IMPORTANT POINTS

- Cannot instantiate an abstract class
- Subclasses must implement all abstract methods, or be abstract themselves
- Abstract classes can have constructors and instance variables

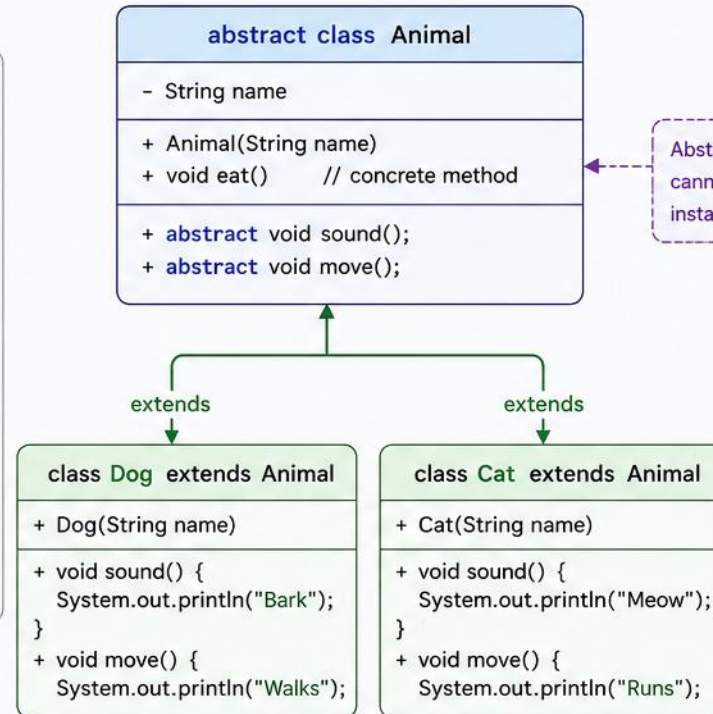
KEY TAKEAWAY Abstract classes define a partial blueprint — subclasses complete the remaining parts.

ABSTRACT CLASSES IN JAVA

An abstract class in Java is a class that cannot be instantiated and is meant to be subclassed. It can have abstract (unimplemented) methods as well as concrete (implemented) methods.

KEY POINTS

- ✓ Declared using the **abstract** keyword.
- ✓ Cannot create objects of an abstract class.
- ✓ Can have **abstract methods** (without body) and **concrete methods** (with body).
- ✓ A subclass must **extend** the abstract class and **implement** all abstract methods.
- ✓ Supports abstraction and provides a common blueprint for subclasses.



EXAMPLE

```
abstract class Animal {
    String name;

    Animal(String name) {
        this.name = name;
    }

    void eat() {
        System.out.println(name + " eats");
    }

    abstract void sound();
    abstract void move();
}

class Dog extends Animal {
    Dog(String name) { super(name); }

    void sound() { System.out.println("Bark"); }
    void move() { System.out.println("Walks"); }
}

class Cat extends Animal {
    Cat(String name) { super(name); }

    void sound() { System.out.println("Meow"); }
    void move() { System.out.println("Runs"); }
}

// Usage
Animal a; // reference
// Animal obj = new Animal("Tom"); // Error

Animal a1 = new Dog("Buddy");
a1.eat(); // Buddy eats
a1.sound(); // Bark
a1.move(); // Walks
```

HOW IT WORKS



Abstract class defines a blueprint with some common behavior and abstract methods.



Subclasses extend the abstract class.



Subclasses provide implementation for all abstract methods.



Objects are created from subclasses and used via superclass reference.

BENEFITS

- ✓ Provides a common structure for related classes.
- ✓ Promotes code reusability and consistency.
- ✓ Achieves abstraction (hide implementation details).
- ✓ Supports the principle of "program to an interface".

ABSTRACT CLASSES IN JAVA

How abstract classes compare with interfaces, and when to use each. (2 of 2)

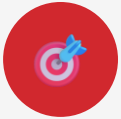
ASPECT	ABSTRACT CLASS	INTERFACE
Methods	Abstract + concrete	Abstract (default) + default + static
Variables	Instance variables	public static final only
Constructor	Yes	No
Inheritance	Single	Multiple
Purpose	"is-a" shared base	Contract a class must implement

Use an abstract class when classes are closely related and share common code. Use an interface when you need a pure contract/capability across unrelated classes, or multiple inheritance of type.

KEY TAKEAWAY Use abstract classes to design a strong foundation for your applications and achieve reusability.

TRY IT YOURSELF — EXERCISE 4: ABSTRACT CLASSES

Reinforces: abstract classes and compiler-enforced contracts, as just covered.



1

Goal

Turn Account into an abstract class with one abstract method



2

File

AbstractAccount.java,
AbstractSavings.java,
AbstractDemo.java



3

Prove it

new AbstractAccount(...)
fails to compile —
abstract classes can't be
instantiated



4

Compile & Run

```
javac *.java && java  
AbstractDemo
```



5

Reference

exercises/exercise-04-
abstract-classes.md

TRY IT NOW Complete Exercise 4 before continuing — see [exercises/exercise-04-abstract-classes.md](#).

INTERFACES IN JAVA

A blueprint of a class: abstract methods (public by default) and constants. (1 of 2)

KEY FEATURES

- Cannot be instantiated
- Contains abstract methods by default (public + abstract)
- Contains constants by default (public static final)
- A class implements an interface with the implements keyword
- A class can implement multiple interfaces
- Supports default and static methods (Java 8+)

```
public interface Drawable {  
    int PI = 3;    // public static final  
  
    void draw();  // abstract  
  
    default void info() {  
        System.out.println("Drawable");  
    }  
  
    static void help() {    // static  
        System.out.println("Help");  
    }  
}
```

? WHY INTERFACES?

- Achieve full abstraction
- Define what a class can do, not how
- Promote loose coupling
- Easy to extend and maintain

KEY TAKEAWAY Interfaces define contracts. A class implementing Drawable must provide its own draw().

INTERFACES IN JAVA

An interface is a blueprint of a class. It contains abstract methods (by default public and abstract) and constants (public, static and final).
A class implements an interface and provides the implementation of its methods.



INTERFACES

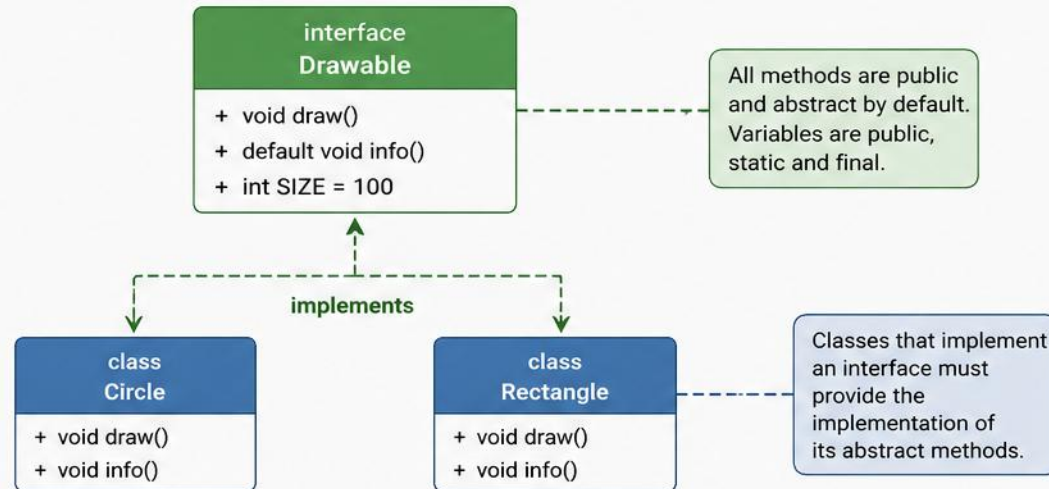
KEY POINTS

- Declared with the keyword **interface**.
- Cannot be instantiated.
- Methods are public and abstract by default.
- Variables are public, **static** and **final** by default.
- A class implements an interface using the keyword **implements**.

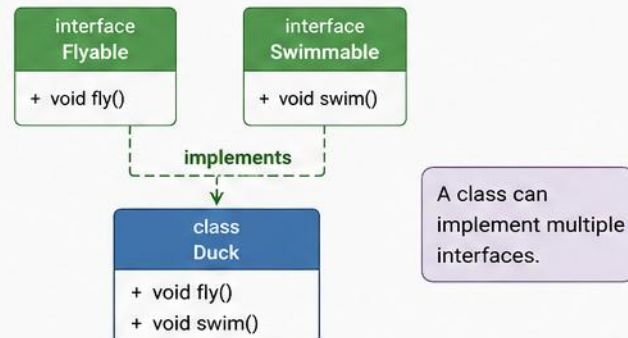
EXAMPLE

```
interface Drawable {  
    void draw();           // abstract method  
    default void info() {  // default method  
        System.out.println("Drawable");  
    }  
    int SIZE = 100;        // public static final  
}  
  
class Circle implements Drawable {  
    public void draw() {  
        System.out.println("Drawing Circle");  
    }  
}  
  
class Rectangle implements Drawable {  
    public void draw() {  
        System.out.println("Drawing Rectangle");  
    }  
}  
  
// Usage  
Drawable d1 = new Circle();  
d1.draw(); // Drawing Circle  
d1.info(); // Drawable
```

HOW INTERFACES WORK



MULTIPLE INHERITANCE WITH INTERFACES



BENEFITS

- ✓ Achieves 100% abstraction.
- ✓ Supports multiple inheritance.
- ✓ Promotes loose coupling.
- ✓ Enhances flexibility and scalability.
- ✓ Used to define common behavior across unrelated classes.

INTERFACES IN JAVA

Implementing an interface, and combining multiple interfaces. (2 of 2)

```
public class Circle implements Drawable {
    private double radius;
    public Circle(double radius) {
        this.radius = radius;
    }
    @Override
    public void draw() {
        System.out.println("Drawing radius " + radius);
    }
}

Drawable d = new Circle(5.0);
d.draw();           // calls Circle's draw()
d.info();           // default method
Drawable.help();    // static method
```

```
interface A { void first(); }
interface B { void second(); }

class MyClass implements A, B {
    public void first() { ...; }
    public void second() { ...; }
}
```

✓ IMPORTANT POINTS

- All interface methods are public + abstract by default
- All interface variables are public static final (constants)
- A class can implement multiple interfaces — multiple inheritance of type

KEY TAKEAWAY A class must provide implementations for all abstract methods of the interfaces it implements.

INTERFACE VS ABSTRACT CLASS IN JAVA

Both achieve abstraction; the difference is purpose, capability, and usage.

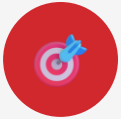
ASPECT	INTERFACE	ABSTRACT CLASS
Methods	Abstract by default; default/static (Java 8+)	Abstract + concrete; can have final methods
Variables	public static final only (constants)	Instance, static, or final variables
Constructors	Cannot have constructors	Can have constructors
Access modifiers	Methods public by default	Any modifier (public/protected/private)
Inheritance	Implement multiple (implements)	Extend only one (extends)
Purpose	Define a contract — what a class can do	Share code & state — what a class is

In short: Interface is for “what a class can do.” Abstract class is for “what a class is.”

KEY TAKEAWAY Use an interface for a capability/contract; use an abstract class to share code and state.

TRY IT YOURSELF — EXERCISE 5: INTERFACE PRACTICE

Reinforces: interfaces and implements, as just covered.



1

Goal

Create interface Printable
{ void printDetails(); } and
implement it



2

File

Printable.java,
Customer.java,
InterfaceDemo.java in
examples/module-03-
exercises/



3

Key concept

implements Printable,
then call printDetails()
through a Printable
reference



4

Compile & Run

```
javac *.java && java  
InterfaceDemo
```



5

Reference

exercises/exercise-05-
interface.md

TRY IT NOW Complete Exercise 5 before continuing — see [exercises/exercise-05-interface.md](#).

INTRODUCTION TO SOLID PRINCIPLES

Five design principles for writing maintainable, flexible, scalable OOP software.

? WHAT IS SOLID?

SOLID is an acronym for five object-oriented design principles introduced by Robert C. Martin ("Uncle Bob"). They help developers create systems that are easier to understand, maintain, extend, and test.

★ WHY SOLID?

- Improves maintainability and enhances scalability
- Reduces impact of changes; easier testing & debugging
- Promotes good design practices

⚖️ THE FIVE PRINCIPLES (S.O.L.I.D.)

- S — Single Responsibility: one reason to change
- O — Open/Closed: open for extension, closed for modification
- L — Liskov Substitution: subclasses substitute for their base class
- I — Interface Segregation: no client depends on unused methods
- D — Dependency Inversion: depend on abstractions, not concretions

KEY TAKEAWAY Following SOLID principles leads to clean, robust, maintainable code that's easier to extend and adapt.

SOLID PRINCIPLES

Five Design Principles for Maintainable, Flexible and Scalable Software

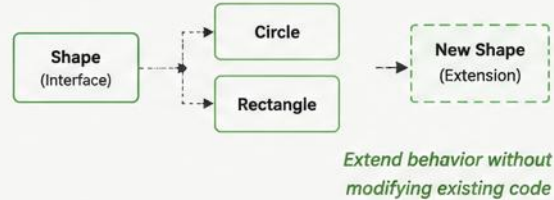
1. SINGLE RESPONSIBILITY PRINCIPLE (SRP)

A class should have only one reason to change, meaning it should have only one job.



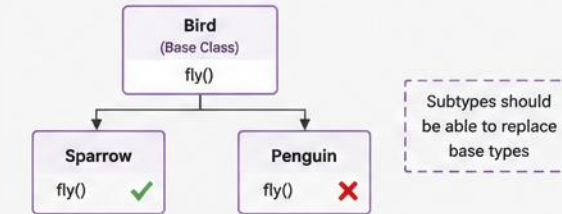
2. OPEN/CLOSED PRINCIPLE (OCP)

Software entities should be open for extension, but closed for modification.



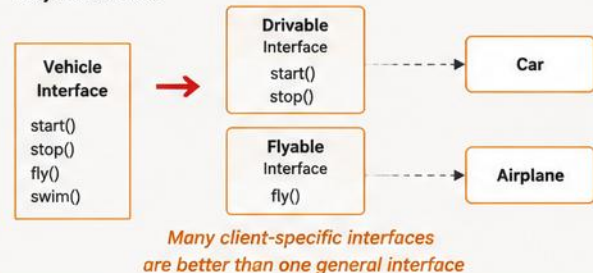
3. LISKOV SUBSTITUTION PRINCIPLE (LSP)

Subtypes must be substitutable for their base types without altering the correctness of the program.



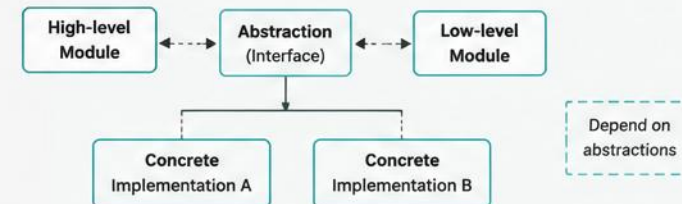
4. INTERFACE SEGREGATION PRINCIPLE (ISP)

Clients should not be forced to depend on interfaces they do not use.



5. DEPENDENCY INVERSION PRINCIPLE (DIP)

Depend on abstractions, not on concretions. High-level modules should not depend on low-level modules. Both should depend on abstractions.



BENEFITS



Improved
Maintainability



Greater
Flexibility



Enhanced
Testability



Reduced
Coupling



Easier
Scalability

THE FIVE SOLID PRINCIPLES

One line per principle — the detail pages that follow expand each one.



Single Responsibility

A class should have only one reason to change.



Open / Closed

Open for extension, closed for modification.



Liskov Substitution

Subtypes must be substitutable for their base types.



Interface Segregation

Clients shouldn't depend on unused interfaces.



Dependency Inversion

Depend on abstractions, not on concretions.

KEY TAKEAWAY SOLID: Single Responsibility, Open/Closed, Liskov Substitution, Interface Segregation, Dependency Inversion.

SINGLE RESPONSIBILITY PRINCIPLE (SRP)

A class should have only one reason to change — only one job or responsibility.

```
// ❌ Bad: one class, two responsibilities
class Employee {
    void setEmployee(int id, String name, double salary) {...}

    // Responsibility 2: generate report
    String generateReport() {
        return "ID:"+id+" Name:"+name;
    }
}
```

```
// ✅ Good: separate responsibilities
class Employee { ...getters... }

class EmployeeReportGenerator {
    String generateReport(Employee e) {
        return "ID:"+e.getId();
    }
}
```

? WHY IS SRP IMPORTANT?

- Easier maintenance — changes are localized
- Fewer bugs — smaller, focused classes are less error-prone
- Better testability — each class can be tested in isolation

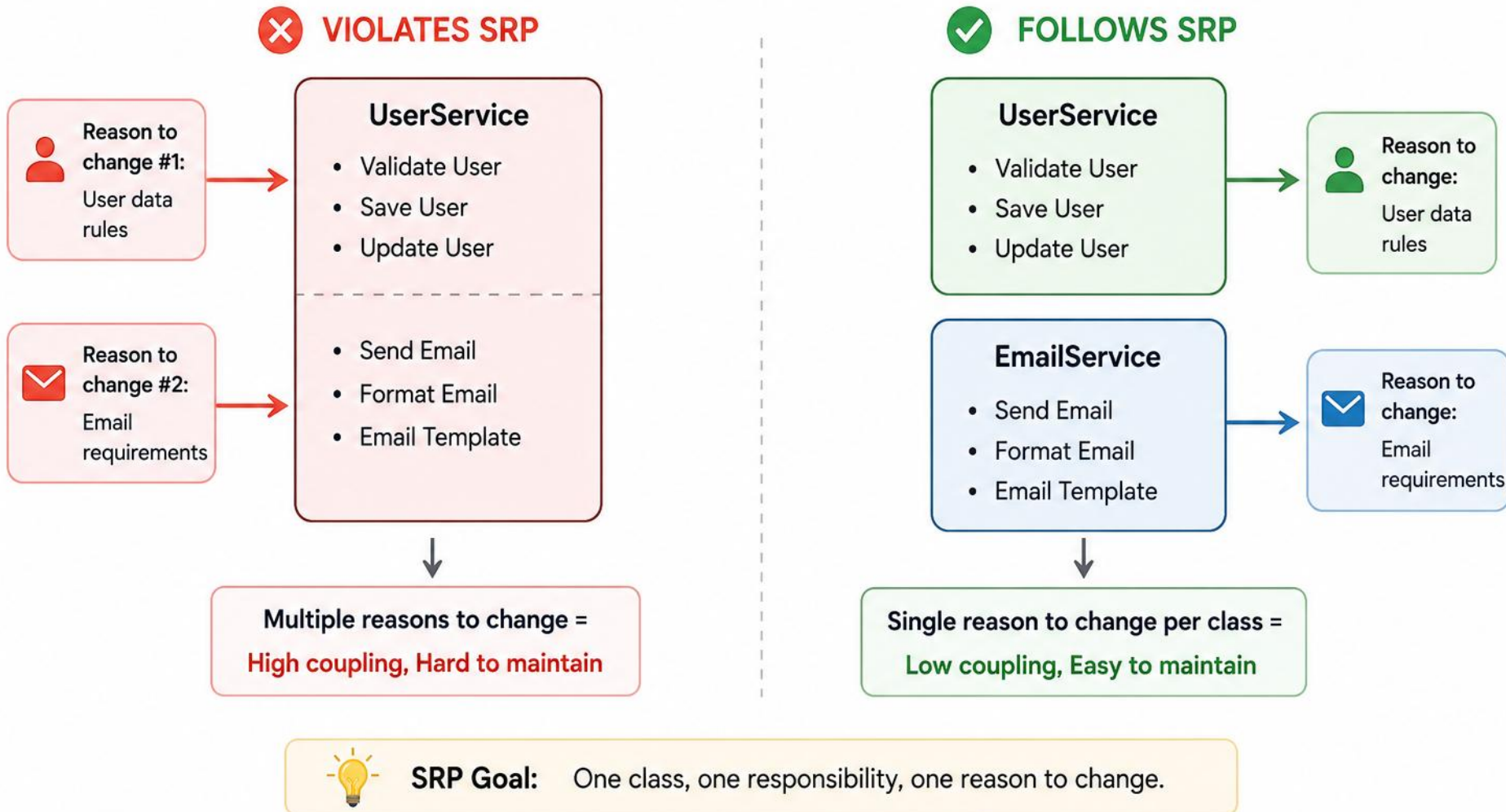
👨‍🍳 REAL-WORLD ANALOGY

A restaurant Chef cooks food; a Waiter takes orders and serves. Each has one job, one reason to change — just like well-designed classes.

KEY TAKEAWAY High cohesion in a class, low coupling between classes — the key to good design.

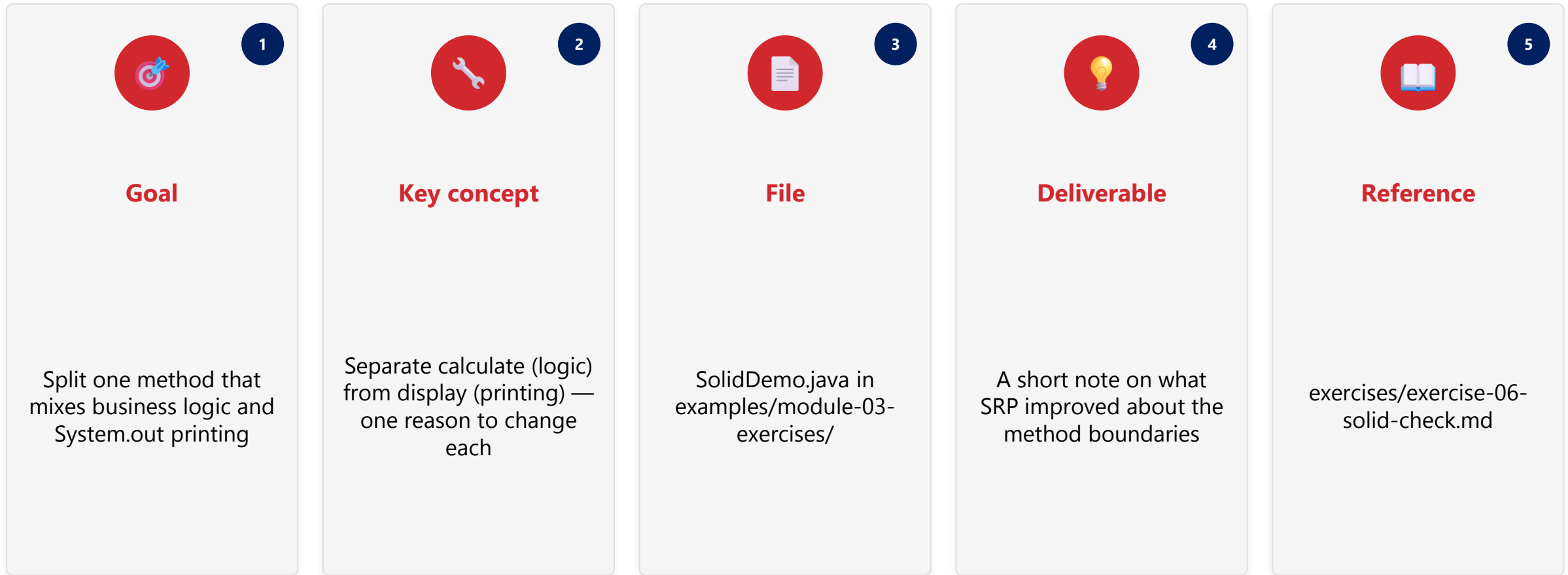
SINGLE RESPONSIBILITY PRINCIPLE

A class should have only one reason to change, meaning it should have only one job or responsibility.



TRY IT YOURSELF — EXERCISE 6: SOLID SPOT-CHECK

Reinforces: the Single Responsibility Principle you just saw.



TRY IT NOW Complete Exercise 6 before continuing — see [exercises/exercise-06-solid-check.md](#).

OPEN/CLOSED PRINCIPLE (OCP)

Software entities should be OPEN for extension, but CLOSED for modification.

```
// ❌ Violates OCP – must edit for every shape
class ShapeAreaCalculator {
    double area(String type, double r,
                double l, double w) {
        if (type.equals("CIRCLE"))
            return 3.14 * r * r;
        // add more if/else for each new shape...
    }
}
```

```
// ✅ Follows OCP – extend, don't modify
interface Shape { double area(); }
class Circle implements Shape {
    double r;
    public double area(){ return 3.14*r*r; }
}
// new shape = new class, zero edits above
```

🔑 KEY POINTS

- Do not modify existing, tested code
- Extend behavior using new classes
- Use abstraction (interfaces / abstract classes)
- Leads to loosely coupled, scalable code

✅ WHEN TO USE OCP?

When requirements are likely to change, features are added frequently, or the system needs long-term maintenance — e.g., in frameworks and libraries.

KEY TAKEAWAY Build systems that can grow without breaking what already works.

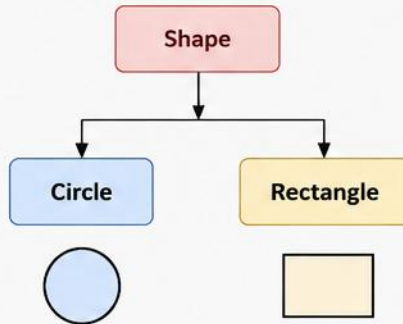
OPEN/CLOSED PRINCIPLE (OCP)

Software entities (classes, modules, functions) should be **open for extension**, but **closed for modification**.

BEFORE: Not Closed for Modification

To add a new shape, we must **modify** existing code (Shape class).

```
class Shape {
    double area(Shape s) {
        if (s instanceof Circle)
            return 3.14 * s.r * s.r;
        else if (s instanceof Rectangle)
            return s.l * s.w;
        // To add new shape,
        // we must change this code!
    }
}
```

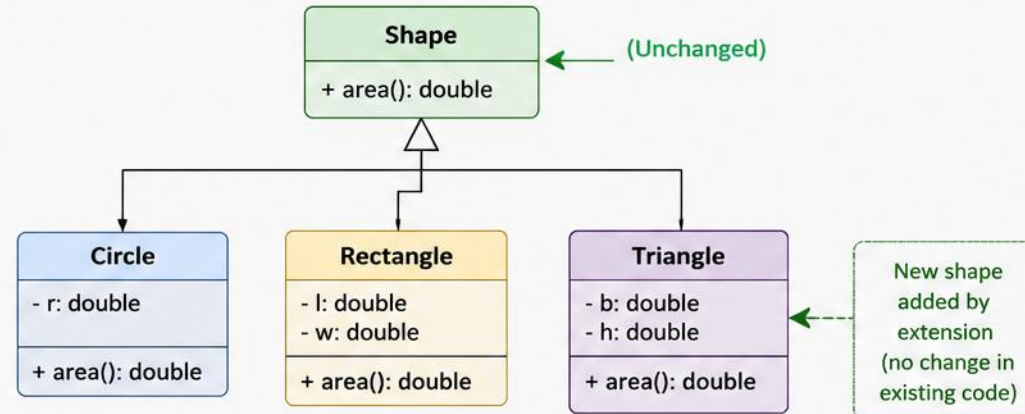


Problem

- ✗ Existing code changes when new shape is added.
- ✗ Higher risk of bugs.
- ✗ Violates OCP.

AFTER: Closed for Modification (Open for Extension)

We **extend** the behavior by adding new classes, **without modifying** existing code.



Client Code (Unchanged)

```
Shape s1 = new Circle(5);
Shape s2 = new Rectangle(4, 6);
Shape s3 = new Triangle(3, 4); // New - no change in existing code
System.out.println(s1.area());
System.out.println(s2.area());
System.out.println(s3.area());
```

Benefits

- ✓ Closed for modification
- ✓ Open for extension
- ✓ Easier to maintain
- ✓ Less risk of bugs
- ✓ Follows OCP



Key Idea: Design your modules so that new functionality can be added by extension (new classes) instead of modifying existing, tested code.

LSKOV SUBSTITUTION PRINCIPLE (LSP)

Objects of a superclass should be replaceable with objects of its subclasses without breaking correctness.

```
// ❌ Violates LSP
class Rectangle {
    void setWidth(int w) { width = w; }
    void setHeight(int h) { height = h; }
}
class Square extends Rectangle {
    void setWidth(int w) { width=height=w; }
} // breaks Rectangle's contract!
```

```
// ✅ Follows LSP – use abstraction
interface Shape { double area(); }
class Rectangle implements Shape {...}
class Square implements Shape {...}
// Both fully substitutable wherever Shape is expected
```

🔑 KEY POINTS

- Subclasses must be substitutable for their base class
- Subclass must honor the contract (behavior) of the base class
- Preserves correctness; prevents unexpected behavior
- A subclass should not break the rules defined by its superclass

🐦 REAL-WORLD ANALOGY

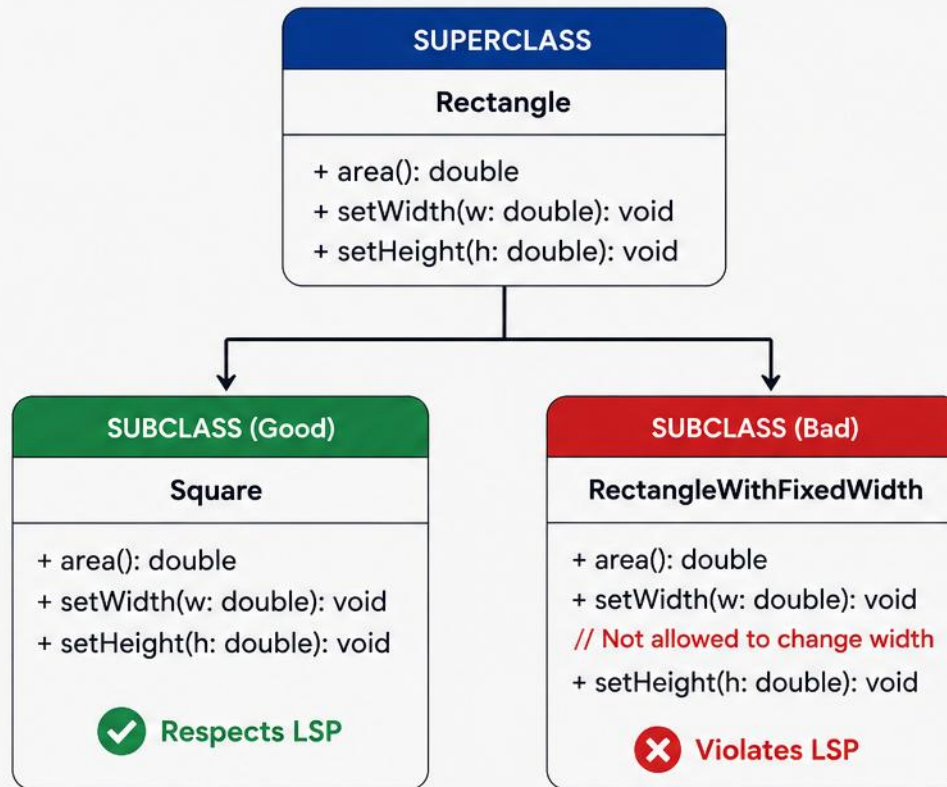
A Bird superclass with fly(): if Penguin extends Bird and can't fly, substitution breaks. Use an abstraction that only defines what all birds truly share.

KEY TAKEAWAY Design subtype hierarchies carefully — when in doubt, prefer composition over inheritance.

LSKOV SUBSTITUTION PRINCIPLE (LSP)



Objects of a subclass should be replaceable with objects of the superclass without affecting the correctness of the program.



USING THE SUPERCLASS

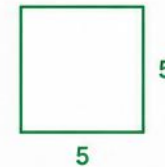
```
void printArea(Rectangle r) {
    r.setWidth(5);
    r.setHeight(4);
    System.out.println(r.area());
}
```

Works
Correctly



```
Rectangle r = new Square();
printArea(r);
```

Output: 20.0



Fails
(Unexpected)



```
Rectangle r = new RectangleWithFixedWidth();
printArea(r);
```

Output: Incorrect!

Width cannot be changed,
behavior breaks.



LSP in Action: A subclass must fully honor the contract of the superclass so that it can be used interchangeably without altering the desired behavior of the program.

INTERFACE SEGREGATION PRINCIPLE (ISP)

Clients should not be forced to depend on interfaces they do not use.

```
// ❌ Fat interface forces unused methods
interface Worker {
    void work(); void eat(); void drive();
}

class HumanWorker implements Worker {
    public void drive() {
        throw new UnsupportedOperationException();
    } // Not all workers can drive!
}
```

```
// ✅ Small, focused interfaces
interface Workable { void work(); }
interface Eatable { void eat(); }
interface Drivable { void drive(); }

class HumanWorker implements
    Workable, Eatable { ...; }
```

🔑 KEY POINTS

- Create small, focused interfaces for specific client needs
- No client should be forced to implement methods it doesn't use
- Reduces coupling between unrelated interfaces and clients
- Improves maintainability and readability

🍷 REAL-WORLD ANALOGY

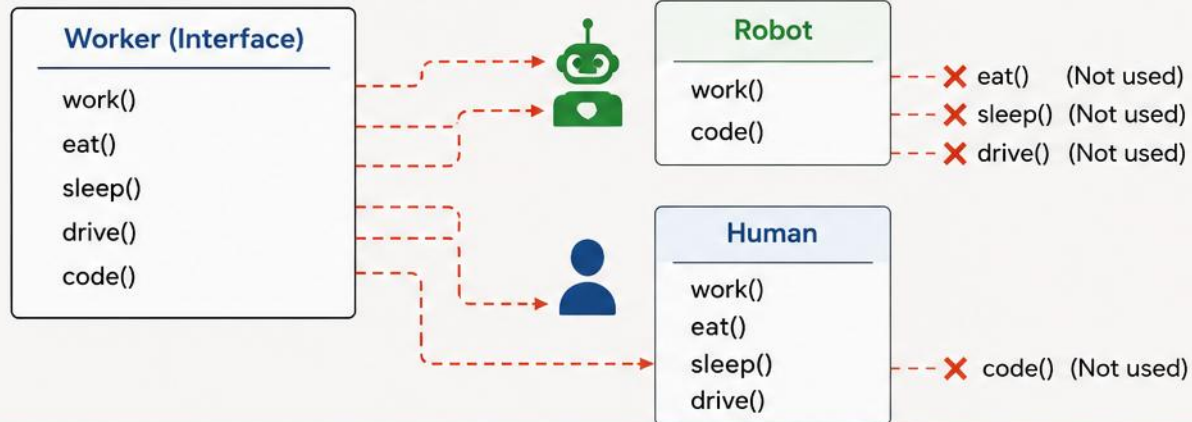
A waiter doesn't need to cook. Give chefs a Cooking interface and waiters a Serving interface — not one giant "restaurant worker" interface.

KEY TAKEAWAY Many small, client-specific interfaces beat one large, general-purpose interface.

INTERFACE SEGREGATION PRINCIPLE (ISP)

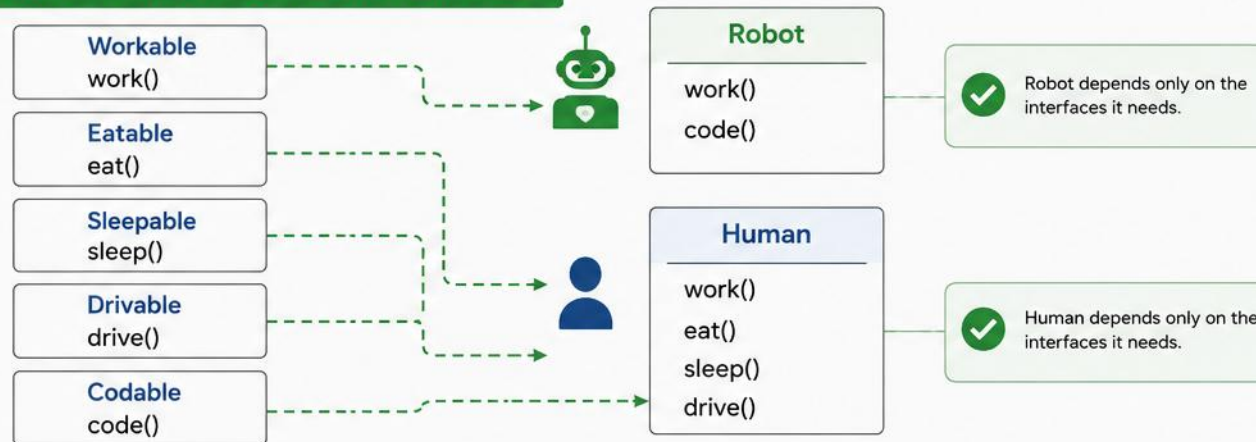
Clients should not be forced to depend on interfaces they do not use.

PROBLEM: FAT INTERFACE



- The Worker interface forces Robot to depend on methods it does not use (eat, sleep, drive).
- Any change in the unused methods can impact the clients unnecessarily.
- Leads to bloated interfaces and tighter coupling.

SOLUTION: SEGREGATED INTERFACES



- Clients are not forced to depend on methods they do not use.
- Interfaces are focused and cohesive.
- Easier to maintain, extend and test.
- Reduces impact of changes and improves flexibility.

---> Depends on (unused methods included)

---> Depends on (only required methods)

DEPENDENCY INVERSION PRINCIPLE (DIP)

High-level modules should not depend on low-level modules. Both should depend on abstractions.

```
// ❌ Tight coupling to a concrete class
class MySQLDatabase { void save(){} }
class UserService {
    private MySQLDatabase db =
        new MySQLDatabase();
    // swapping DB means editing this class
}
```

```
// ✅ Depend on an abstraction
interface Database { void save(String d); }
class MySQLDatabase implements Database {...}
class UserService {
    private Database db;
    UserService(Database db){ this.db=db; }
}
```

🔑 KEY POINTS

- Depend on abstractions, not concrete classes
- High-level and low-level modules both depend on abstractions
- Reduces coupling between components
- Use Dependency Injection to achieve DIP

💡 REAL-WORLD ANALOGY

A light switch doesn't depend on one specific bulb — it depends on an electrical interface. Any bulb (LED, CFL) can be plugged in.

KEY TAKEAWAY Program to an interface, not to an implementation. Abstractions are stable; details can change.

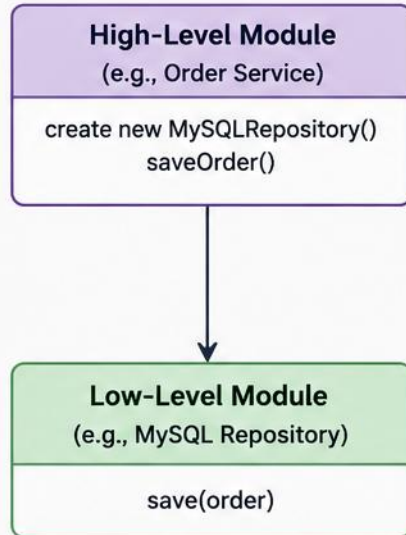
DEPENDENCY INVERSION PRINCIPLE (DIP)

High-level modules should not depend on low-level modules. Both should depend on abstractions.

Abstractions should not depend on details. Details should depend on abstractions.

WITHOUT DIP (Violation)

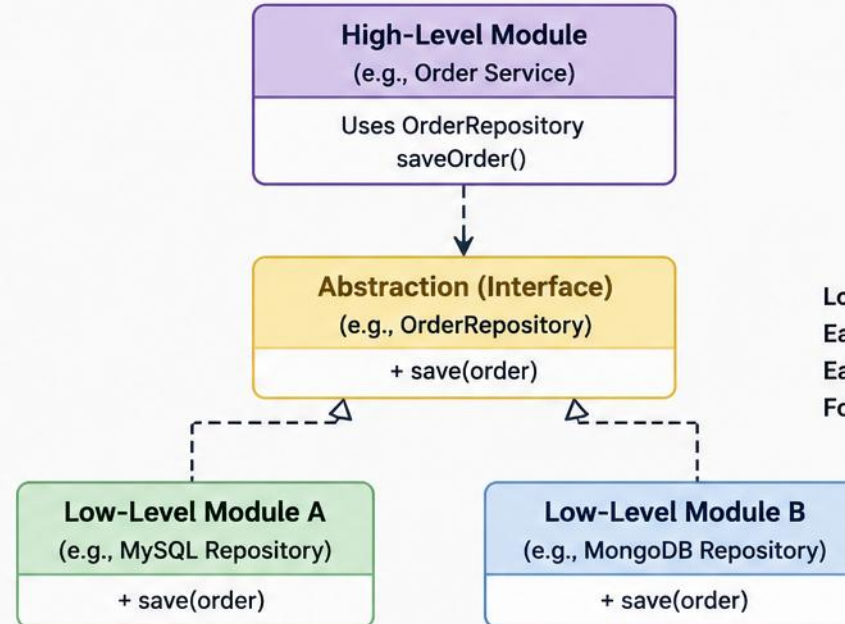
High-level module depends on low-level module



Tight coupling
Hard to change
Hard to test
Violates DIP

WITH DIP (Compliant)

Both high-level and low-level modules depend on abstraction



Loose coupling
Easy to change
Easy to test
Follows DIP



Key Takeaway

Depend on abstractions,
not on concrete implementations.

Benefits



Reduced
Coupling



Greater
Flexibility



Easier
Testing



Easier to Extend
and Maintain

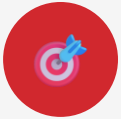
Legend

- - -> Depends on (abstraction)

- - > Implements (realizes)

TRY IT YOURSELF — EXERCISE 7: SOLID BEYOND SRP

Reinforces: OCP, LSP, ISP, and DIP, as just covered.



1

Goal

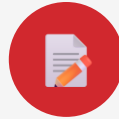
Spot-check the other four SOLID principles using the Account hierarchy



2

New file

FrozenAccount.java — an LSP-safe subclass with no special-casing



3

Write

One sentence each for OCP, LSP, ISP, and DIP



4

Key concept

A subclass must honor its parent's contract, not just compile



5

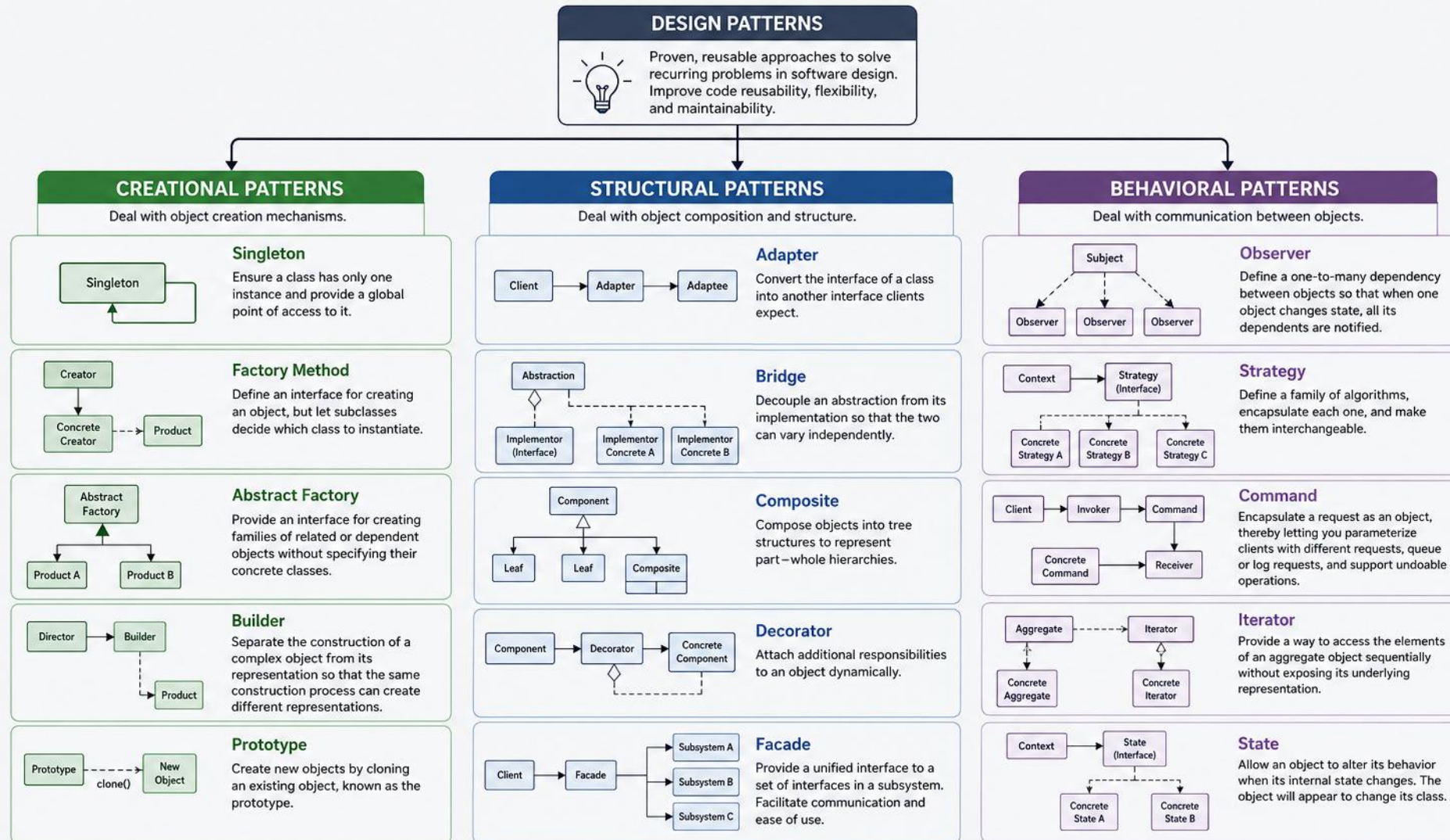
Reference

[exercises/exercise-07-solid-beyond-srp.md](#)

TRY IT NOW Complete Exercise 7 before continuing — see [exercises/exercise-07-solid-beyond-srp.md](#).

OOP DESIGN PATTERNS

Reusable solutions to common object-oriented design problems.



BENEFITS

• Reusability • Maintainability • Flexibility • Scalability • Best Practices



PRINCIPLE

Program to an interface, not an implementation.

OOP DESIGN PATTERNS IN ENTERPRISE APPLICATIONS

Proven, reusable solutions to common design problems. (1 of 4 — Creational)



Creational — Object creation



Structural — Object composition



Behavioral — Communication & responsibility

Singleton

One instance, global access point.

USE IN ENTERPRISE

Logging, Config, DB connections

Factory Method

Lets subclasses decide which class to instantiate.

USE IN ENTERPRISE

Plugins, Notification systems

Abstract Factory

Creates families of related objects.

USE IN ENTERPRISE

Cross-platform UI toolkits

Builder

Separates complex object construction from representation.

USE IN ENTERPRISE

Reports, Documents, Queries

Prototype

Creates new objects by cloning a prototype.

USE IN ENTERPRISE

Templates, Caching

KEY TAKEAWAY Creational patterns deal with object creation mechanisms — flexible, decoupled ways to build objects.

OOP DESIGN PATTERNS IN ENTERPRISE APPLICATIONS

Structural patterns deal with object composition and structure. (2 of 4)

Adapter

Lets incompatible interfaces work together.

Use: 3rd-party libraries, Legacy systems

Facade

Simplified interface to a complex subsystem.

Use: Complex modules, Layered architectures

Decorator

Adds responsibilities to an object dynamically.

Use: I/O streams, Feature augmentation

Composite

Tree structures for part-whole hierarchies.

Use: UI components, File systems, Orgs

Flyweight

Shares fine-grained objects efficiently.

Use: Memory optimization, caches

Proxy

Placeholder that controls access to an object.

Use: Security, Lazy loading, Remote services

KEY TAKEAWAY Structural patterns compose classes and objects into larger, flexible structures without tight coupling.

OOP DESIGN PATTERNS IN ENTERPRISE APPLICATIONS

Behavioral patterns deal with object communication and responsibility. (3 of 4)

Strategy

Encapsulates interchangeable algorithms.

USE IN ENTERPRISE

Payment methods, Pricing, Sorting

Observer

One-to-many notification when state changes.

USE IN ENTERPRISE

Event handling, Notifications

Command

Encapsulates a request as an object.

USE IN ENTERPRISE

Undo/Redo, Job queues,
Transactions

Template Method

Defines algorithm skeleton; subclasses fill steps.

USE IN ENTERPRISE

Report generation, Import/Export

KEY TAKEAWAY Behavioral patterns govern how objects communicate and share responsibility. (Part 1 of 2 — 3 more follow.)

OOP DESIGN PATTERNS IN ENTERPRISE APPLICATIONS

Behavioral patterns, continued. (4 of 4)

State

Changes behavior when internal state changes.

USE IN ENTERPRISE

Order processing, Workflow, Game states

Chain of Responsibility

Passes a request along a chain of handlers.

USE IN ENTERPRISE

Approval workflows, Exception handling

Mediator

Encapsulates how a set of objects interact.

USE IN ENTERPRISE

Chat systems, GUI components

KEY TAKEAWAY Choose patterns that add real value — don't overuse them (YAGNI). Prefer simplicity when possible.

MODELING REAL-WORLD BUSINESS ENTITIES

Accurate models of business entities are the foundation of maintainable enterprise apps. (1 of 2)

? WHAT IS AN ENTITY?

A distinct, identifiable object or concept in the business domain that has a lifecycle and holds important data.

- Has a unique identity
- Has attributes (data)
- Has behavior (operations)
- Has relationships with other entities

EXAMPLE: E-COMMERCE ORDER



Customer

Places orders



Order

Contains order items



Payment

Payment for an order



Shipment

Delivery details

KEY MODELING GUIDELINES

- Identify entities — use nouns from business language
- Define clear responsibilities for each entity
- Use value objects for descriptive, immutable data
- Model relationships & multiplicities explicitly
- Enforce business rules close to the data

KEY TAKEAWAY Good domain models capture the heart of your business — they make code easier to understand and extend.

MODELING REAL-WORLD BUSINESS ENTITIES IN JAVA

Represent business concepts as classes and define their attributes, behaviors, and relationships.

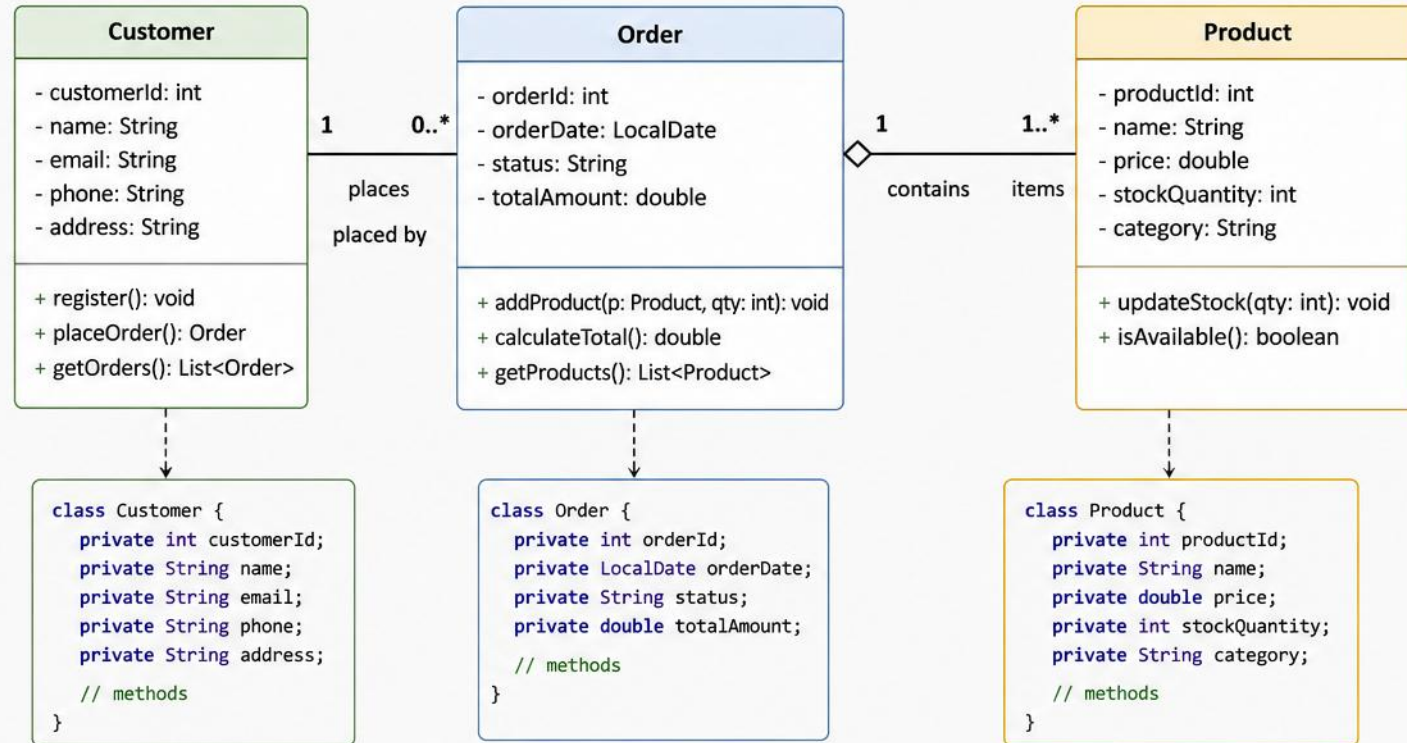
REAL-WORLD SCENARIO



An online store manages **Customers** who place **Orders**. Each Order contains one or more **Products**.

HOW WE MODEL IT

- 1 Identify Entities → Customer, Order, Product
- 2 Define Attributes → Properties of each entity
- 3 Define Behaviors → Actions each entity can perform
- 4 Define Relationships → How entities are associated



UML NOTATION GUIDE



Class
Represents an entity
- attribute: Type
Private attribute
+ method(): Type
Public method



Association
Relationship between entities

1 One
0..* Zero or many



Aggregation
Whole-part relationship
(Order contains Products)

BENEFITS

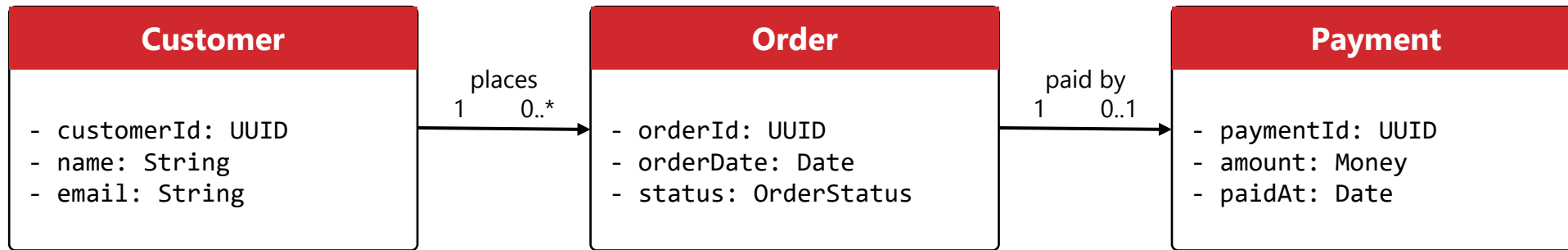
- ✓ Maps real-world concepts to code
- ✓ Creates a clear structure
- ✓ Improves maintainability & scalability
- ✓ Facilitates communication between business & technical teams



Good modeling is the foundation of good software design.

DOMAIN MODEL: UML CLASS DIAGRAM

Customer, Order, and Payment — modeled with associations and multiplicities. (2 of 2)



DESIGN NOTES

- A Customer can place many Orders (1-to-many)
- An Order is paid by zero or one Payment (0..1, since payment is optional until an order is confirmed)
- Payment stores the amount and paidAt timestamp for that order



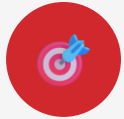
BENEFITS IN ENTERPRISE APPS

- Shared understanding — ubiquitous language across teams
- Maintainability — clear models reduce complexity
- Data integrity — business rules protected in the model

KEY TAKEAWAY Model relationships and multiplicities explicitly; keep the model simple and evolve it iteratively.

TRY IT YOURSELF — EXERCISE 8: MINI UML

Reinforces: the class diagram notation you just saw.



1

Goal

Draw Account, SavingsAccount, and Customer with their relationships



2

Deliverable

Paper or Mermaid diagram — no code required



3

Show

Inheritance (SavingsAccount → Account) and association (Customer → Account)



4

Key concept

Note multiplicities — one Customer can have many Accounts (1 : many)



5

Reference

[exercises/exercise-08-uml-mini.md](#)

TRY IT NOW Complete Exercise 8 before continuing — see [exercises/exercise-08-uml-mini.md](#).

PRE-LAB EXERCISES OVERVIEW

Eight short warm-ups to practice before the full Lab 3 build.

PURPOSE

Eight short, focused exercises that let you practice one OOP concept at a time — entities, encapsulation, inheritance, abstract classes, interfaces, UML, and SOLID — using a simple banking domain, before you start the full Lab 3 build.

HOW TO USE THEM

- Work through them in order, after the module slides
- Keep practice code separate from the graded Lab 3 submission
- Each one takes minutes, not hours

THE EIGHT EXERCISES

Located in labs/module-03/exercises/ — complete all eight before opening the full Lab 3 guide.

- **Exercise 1** — List entities and one responsibility each
- **Exercise 2** — Private balance with deposit/withdraw
- **Exercise 3** — Override withdrawal across two subclasses
- **Exercise 4** — Make Account abstract with one method
- **Exercise 5** — Define and implement a simple interface
- **Exercise 6** — Split calculation from display logic
- **Exercise 7** — Spot-check the other four principles
- **Exercise 8** — Sketch a small class diagram

READY FOR LAB 3?

- Each exercise compiles and runs (or notes are complete for analysis-only tasks)
- You can explain each result in one sentence
- Practice files are kept separate from the Lab 3 submission
- You completed all eight exercises in order
- You are comfortable with entities, encapsulation, and inheritance
- You are ready to open the full Lab 3 guide

KEY TAKEAWAY Eight quick exercises build the muscle memory you will need for the full Lab 3 Banking Management System build.

LAB OVERVIEW: BANKING MANAGEMENT SYSTEM

Build a menu-driven Banking Management System that applies core OOP pillars.

LAB OBJECTIVE

Build a menu-driven Banking Management System (package `com.academy.bank`) that applies encapsulation, inheritance, abstraction, interfaces, and polymorphism using plain JDK — no frameworks, no database.

YOU WILL PRACTICE

- Abstract class hierarchy: Account → Savings/Current
- Interface contract: Printable
- Polymorphic display via an `Account[]` array

BUSINESS DOMAIN

Bank staff need a console tool to create customers, open Savings or Current accounts, deposit and withdraw funds, and display balances — no database, just plain JDK.

- **Customer** — Profile; implements Printable
- **Account** — Balance, deposit/withdraw
- **Savings** — Interest + display
- **Current** — Fee + display
- **Transaction** — Deposit/withdraw log
- **Service** — Arrays + operations

KEY REQUIREMENTS

- Create customers and open Savings or Current accounts
- Deposit and withdraw with balance validation
- Display accounts and customers polymorphically
- Keep Main thin — operations live in `BankService`
- Compile with `javac -d out` and run with `java -cp out`
- Sketch a UML class diagram of the design

KEY TAKEAWAY Build a menu-driven Banking Management System — encapsulation, inheritance, interfaces, and polymorphism working together.

LAB TASKS AND DELIVERABLES

Build the Account hierarchy, service layer, and menu for the Banking Management System.

LAB TASKS (IN SEQUENCE)

- 1 Create Project & Package — set up the com.academy.bank folder structure
- 2 Define Printable & Customer — interface contract and customer profile
- 3 Build Abstract Account — shared fields, deposit()/withdraw(), abstract displayAccount()
- 4 Create SavingsAccount & CurrentAccount — interest vs. transaction fee, overridden display
- 5 Add Transaction — record deposit/withdraw activity for each account
- 6 Build BankService — arrays plus create/deposit/withdraw/display operations
- 7 Create Menu-Driven Main — thin entry point that delegates to BankService
- 8 Compile, Run & Document — verify polymorphic display, sketch UML, apply SOLID checklist

EXPECTED DELIVERABLES

- **Source Code**
Full com.academy.bank sources: Customer, Account, Savings, Current, Printable, Transaction, BankService, Main
- **Screenshots**
Create customer/account, deposit, withdraw, polymorphic display, exit
- **UML Class Diagram**
Inheritance, interface, and “uses” relationships
- **Design Note**
SOLID principles and inheritance/polymorphism explained in your own words
- **Compile/Run Evidence**
javac -d out and java -cp out commands with output

KEY TAKEAWAY 100-mark rubric: class design 20, inheritance/interface 25, polymorphism/service 25, menu/compile 15, UML/quality/evidence 15.

MODULE 3 SUMMARY — WHAT WE ACHIEVED

We learned to design real-world software using OOP principles, SOLID, and design patterns.



Understand

Real-world business entities and their behavior.



Design

Robust and maintainable class models.



Implement

Clean, modular and extensible code using OOP.



Apply

Design patterns and SOLID principles for better solutions.



OOP DESIGN PROCESS

- 1 Analyze Requirements
- 2 Identify Entities
- 3 Model Relationships
- 4 Design Class Diagram
- 5 Apply SOLID & Patterns
- 6 Implement & Test
- 7 Review & Improve



KEY TAKEAWAYS

- Good design starts with understanding the business
- Clear models lead to better code and fewer defects
- SOLID principles guide flexible, maintainable software
- Design patterns provide tested solutions to common problems
- Continuous improvement and refactoring keep systems healthy

KEY TAKEAWAY Strong design leads to systems that are easy to extend, test, understand — and aligned with business goals.

MODULE 3 KEY TAKEAWAYS

What we learned, and what this module produced. (1 of 2)

WHAT WE LEARNED

- Modeling business entities — attributes, behaviors, responsibilities
- OOP principles — encapsulation, inheritance, polymorphism, abstraction
- Interfaces and abstract classes for full and partial abstraction
- SOLID principles (SRP, OCP, LSP, ISP, DIP) for maintainable design
- Creational, Structural, and Behavioral design patterns
- UML class diagrams to visualize structure and relationships

KEY OUTPUTS FROM THIS MODULE

- Class designs modeling Customer, Account, Transaction, Order
- Encapsulated fields with validated getters/setters
- Inheritance hierarchies and interface-based contracts
- SOLID-compliant, extensible class structures
- UML class diagram and design note for the lab project
- Working Java programs demonstrating each OOP pillar

KEY TAKEAWAY Clear models help everyone — developers, testers, and business — stay aligned.

MODULE 3 KEY TAKEAWAYS

The impact of good OOP design, and the SOLID principles at a glance. (2 of 2)



Better Understanding

Clear models keep everyone aligned.



High Maintainability

Good design reduces complexity & technical debt.



Extensibility

Open/Closed principle allows easy extension.



Reusability

Well-designed components can be reused.



THE SOLID PRINCIPLES AT A GLANCE



Single Responsibility



Open / Closed



Liskov Substitution



Interface Segregation



Dependency Inversion

KEY TAKEAWAY Good design is not about writing more code — it's about writing the right code.

MODULE 3 KNOWLEDGE CHECK

Test your understanding of key concepts from Module 3. (1 of 2)

1

Which is the best definition of an entity?

☒ A distinct object/concept with a lifecycle and data

2

Which UML diagram shows classes, attributes, methods & relationships?

☒ Class Diagram

3

In UML, which symbol represents composition?

☒ Filled diamond

4

Which SOLID principle: a class should have one reason to change?

☒ Single Responsibility Principle

5

Which pattern provides a simplified interface to a complex subsystem?

☒ Facade

KEY TAKEAWAY Choose the best answer for each question — questions 6–10 continue on the next slide.

MODULE 3 KNOWLEDGE CHECK

Test your understanding of key concepts from Module 3. (2 of 2)

6

Which principle lets you add functionality without modifying existing code?

✓ **Open/Closed Principle**

7

Customer places many Orders — which relationship describes this?

✓ **Association (1 to Many)**

8

Which principle: clients shouldn't depend on interfaces they don't use?

✓ **Interface Segregation Principle**

9

Which pattern notifies dependents automatically when one object changes?

✓ **Observer**

10

Primary focus when designing class models for enterprise apps?

✓ **Reflect the real-world business domain accurately**

SCORING GUIDE

9–10: Excellent • 7–8: Good • 5–6: Needs Improvement • Below 5: Review the module again

KEY TAKEAWAY Practice, apply, and refactor your designs. Great design leads to great software.

Nicely done

You can design, build, and reason about object-oriented Java programs.

SOLID principles, design patterns, and real domain modeling are now part of your toolkit.

